

every element of S_n can be written in many different ways as a (nondisjoint) product of cycles. For example, even in S_3 the element $\sigma = (1\ 2\ 3)$ may be written

$$\sigma = (1\ 2\ 3) = (1\ 3)(1\ 2) = (1\ 2)(1\ 3)(1\ 2)(1\ 3) = (1\ 2)(2\ 3)$$

and, in fact, there are an infinite number of different ways to write σ . Not requiring the cycles to be disjoint totally destroys the uniqueness of a representation of a permutation as a product of cycles. We can, however, obtain a sort of “parity check” from writing permutations (nonuniquely) as products of 2-cycles.

Definition. A 2-cycle is called a *transposition*.

Intuitively, every permutation of $\{1, 2, \dots, n\}$ can be realized by a succession of transpositions or simple interchanges of pairs of elements (try this on a small deck of cards sometime!). We illustrate how this may be done. First observe that

$$(a_1\ a_2\ \dots\ a_m) = (a_1\ a_m)(a_1\ a_{m-1})(a_1\ a_{m-2})\ \dots\ (a_1\ a_2)$$

for any m -cycle. Now any permutation in S_n may be written as a product of cycles (for instance, its cycle decomposition). Writing each of these cycles in turn as a product of transpositions by the above procedure we see that

every element of S_n may be written as a product of transpositions

or, equivalently,

$$S_n = \langle T \rangle \quad \text{where} \quad T = \{(i\ j) \mid 1 \leq i < j \leq n\}.$$

For example, the permutation σ in Section 1.3 may be written

$$\begin{aligned} \sigma &= (1\ 12\ 8\ 104)(2\ 13)(5\ 11\ 7)(6\ 9) \\ &= (1\ 4)(1\ 10)(1\ 8)(1\ 12)(2\ 13)(5\ 7)(5\ 11)(6\ 9). \end{aligned}$$

The Alternating Group

Again we emphasize that for any $\sigma \in S_n$ there may be many ways of writing σ as a product of transpositions. For fixed σ we now show that the parity (i.e., an odd or even number of terms) is the same for any product of transpositions equaling σ .

Let $x_1, \dots, x_n$ be independent variables and let Δ be the polynomial

$$\Delta = \prod_{1 \leq i < j \leq n} (x_i - x_j),$$

i.e., the product of all the terms $x_i - x_j$ for $i < j$. For example, when $n = 4$,

$$\Delta = (x_1 - x_2)(x_1 - x_3)(x_1 - x_4)(x_2 - x_3)(x_2 - x_4)(x_3 - x_4).$$

For each $\sigma \in S_n$ let σ act on Δ by permuting the variables in the same way it permutes their indices:

$$\sigma(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\sigma(i)} - x_{\sigma(j)}).$$

For example, if $n = 4$ and $\sigma = (1\ 2\ 3\ 4)$ then

$$\sigma(\Delta) = (x_2 - x_3)(x_2 - x_4)(x_2 - x_1)(x_3 - x_4)(x_3 - x_1)(x_4 - x_1)$$

(we have written the factors in the same order as above and applied σ to each factor to get $\sigma(\Delta)$). Note (in general) that Δ contains one factor $x_i - x_j$ for all $i < j$, and since σ is a bijection of the indices, $\sigma(\Delta)$ must contain either $x_i - x_j$ or $x_j - x_i$, but not both (and certainly no $x_i - x_i$ terms), for all $i < j$. If $\sigma(\Delta)$ has a factor $x_j - x_i$ where $j > i$, write this term as $-(x_i - x_j)$. Collecting all the changes in sign together we see that Δ and $\sigma(\Delta)$ have the same factors up to a product of -1 's, i.e.,

$$\sigma(\Delta) = \pm \Delta, \quad \text{for all } \sigma \in S_n.$$

For each $\sigma \in S_n$ let

$$\epsilon(\sigma) = \begin{cases} +1, & \text{if } \sigma(\Delta) = \Delta \\ -1, & \text{if } \sigma(\Delta) = -\Delta. \end{cases}$$

In the example above with $n = 4$ and $\sigma = (1\ 2\ 3\ 4)$, there are exactly 3 factors of the form $x_j - x_i$ where $j > i$ in $\sigma(\Delta)$, each of which contributes a factor of -1 . Hence

$$(1\ 2\ 3\ 4)(\Delta) = (-1)^3(\Delta) = -\Delta,$$

so

$$\epsilon((1\ 2\ 3\ 4)) = -1.$$

Definition.

- (1) $\epsilon(\sigma)$ is called the *sign* of σ .
- (2) σ is called an *even permutation* if $\epsilon(\sigma) = 1$ and an *odd permutation* if $\epsilon(\sigma) = -1$

The next result shows that the sign of a permutation defines a homomorphism.

Proposition 23. The map $\epsilon : S_n \rightarrow \{\pm 1\}$ is a homomorphism (where $\{\pm 1\}$ is a multiplicative version of the cyclic group of order 2).

Proof: By definition,

$$(\tau\sigma)(\Delta) = \prod_{1 \leq i < j \leq n} (x_{\tau\sigma(i)} - x_{\tau\sigma(j)}).$$

Suppose that $\sigma(\Delta)$ has exactly k factors of the form $x_j - x_i$ with $j > i$, that is $\epsilon(\sigma) = (-1)^k$. When calculating $(\tau\sigma)(\Delta)$, after first applying σ to the indices we see that $(\tau\sigma)(\Delta)$ has exactly k factors of the form $x_{\tau(j)} - x_{\tau(i)}$ with $j > i$. Interchanging the order of the terms in these k factors introduces the sign change $(-1)^k = \epsilon(\sigma)$, and now all factors of $(\tau\sigma)(\Delta)$ are of the form $x_{\tau(p)} - x_{\tau(q)}$, with $p < q$. Thus

$$(\tau\sigma)(\Delta) = \epsilon(\sigma) \prod_{1 \leq p < q \leq n} (x_{\tau(p)} - x_{\tau(q)}).$$

Since by definition of ϵ

$$\prod_{1 \leq p < q \leq n} (x_{\tau(p)} - x_{\tau(q)}) = \epsilon(\tau)\Delta$$

we have $(\tau\sigma)(\Delta) = \epsilon(\sigma)\epsilon(\tau)\Delta$. Thus $\epsilon(\tau\sigma) = \epsilon(\sigma)\epsilon(\tau) = \epsilon(\tau)\epsilon(\sigma)$, as claimed.

To see the proof in action, let $n = 4$, $\sigma = (1\ 2\ 3\ 4)$, $\tau = (4\ 2\ 3)$ so $\tau\sigma = (1\ 3\ 2\ 4)$. By definition (using the explicit Δ in this case),

$$\begin{aligned}(\tau\sigma)(\Delta) &= (1\ 3\ 2\ 4)(\Delta) \\&= (x_3 - x_4)(x_3 - x_2)(x_3 - x_1)(x_4 - x_2)(x_4 - x_1)(x_2 - x_1) \\&= (-1)^5 \Delta\end{aligned}$$

where all factors except the first one are flipped to recover Δ . This shows $\epsilon(\tau\sigma) = -1$. On the other hand, since we already computed $\sigma(\Delta)$

$$\begin{aligned}(\tau\sigma)(\Delta) &= \tau(\sigma(\Delta)) \\&= (x_{\tau(2)} - x_{\tau(3)})(x_{\tau(2)} - x_{\tau(4)})(x_{\tau(2)} - x_{\tau(1)})(x_{\tau(3)} - x_{\tau(4)}) \times \\&\quad \times (x_{\tau(3)} - x_{\tau(1)})(x_{\tau(4)} - x_{\tau(1)}) \\&= (-1)^3 \prod_{1 \leq p < q \leq 4} (x_{\tau(p)} - x_{\tau(q)}) = (-1)^3 \tau(\Delta)\end{aligned}$$

where here the third, fifth, and sixth factors need to have their terms interchanged in order to put all factors in the form $x_{\tau(p)} - x_{\tau(q)}$ with $p < q$. We already calculated that $\epsilon(\sigma) = (-1)^3 = -1$ and, by the same method, it is easy to see that $\epsilon(\tau) = (-1)^2 = 1$ so $\epsilon(\tau\sigma) = -1 = \epsilon(\tau)\epsilon(\sigma)$.

The next step is to compute $\epsilon((i\ j))$, for any transposition $(i\ j)$. Rather than compute this directly for arbitrary i and j we do it first for $i = 1$ and $j = 2$ and reduce the general case to this. It is clear that applying $(1\ 2)$ to Δ (regardless of what n is) will flip exactly one factor, namely $x_1 - x_2$; thus $\epsilon((1\ 2)) = -1$. Now for any transposition $(i\ j)$ let λ be the permutation which interchanges 1 and i , interchanges 2 and j , and leaves all other numbers fixed (if $i = 1$ or $j = 2$, λ fixes i or j , respectively). Then it is easy to see that $(i\ j) = \lambda(1\ 2)\lambda$ (compute what the right hand side does to any $k \in \{1, 2, \dots, n\}$). Since ϵ is a homomorphism we obtain

$$\begin{aligned}\epsilon((i\ j)) &= \epsilon(\lambda(1\ 2)\lambda) \\&= \epsilon(\lambda)\epsilon((1\ 2))\epsilon(\lambda) \\&= (-1)\epsilon(\lambda)^2 \\&= -1.\end{aligned}$$

This proves

Proposition 24. Transpositions are all odd permutations and ϵ is a surjective homomorphism.

Definition. The *alternating group of degree n* , denoted by A_n , is the kernel of the homomorphism ϵ (i.e., the set of even permutations).

Note that by the First Isomorphism Theorem $S_n/A_n \cong \epsilon(S_n) = \{\pm 1\}$, so that the order of A_n is easily determined: $|A_n| = \frac{1}{2}|S_n| = \frac{1}{2}(n!)$. Also, $S_n - A_n$ is the coset of

A_n which is not the identity coset and this is the set of all odd permutations. The signs of permutations obey the usual $\mathbb{Z}/2\mathbb{Z}$ laws:

$$(even)(even) = (odd)(odd) = even$$

$$(even)(odd) = (odd)(even) = odd.$$

Moreover, since ϵ is a homomorphism and every $\sigma \in S_n$ is a product of transpositions, say $\sigma = \tau_1 \tau_2 \cdots \tau_k$, then $\epsilon(\sigma) = \epsilon(\tau_1) \cdots \epsilon(\tau_k)$; since $\epsilon(\tau_i) = -1$, for $i = 1, 2, \dots, k$, $\epsilon(\sigma) = (-1)^k$. Thus the class of $k \pmod{2}$, i.e., the parity of the number of transpositions in the product, is the same no matter how we write σ as a product of transpositions:

$$\epsilon(\sigma) = \begin{cases} +1, & \text{if } \sigma \text{ is a product of an even number of transpositions} \\ -1, & \text{if } \sigma \text{ is a product of an odd number of transpositions.} \end{cases}$$

Finally we give a quick way of computing $\epsilon(\sigma)$ from the cycle decomposition of σ . Recall that an m -cycle may be written as a product of $m - 1$ transpositions. Thus

an m -cycle is an odd permutation if and only if m is even.

For any permutation σ let $\alpha_1 \alpha_2 \cdots \alpha_k$ be its cycle decomposition. Then $\epsilon(\sigma)$ is given by $\epsilon(\alpha_1) \cdots \epsilon(\alpha_k)$ and $\epsilon(\alpha_i) = -1$ if and only if the length of α_i is even. It follows that for $\epsilon(\sigma)$ to be -1 the product of the $\epsilon(\alpha_i)$'s must contain an odd number of factors of (-1) . We summarize this in the following proposition:

Proposition 25. The permutation σ is odd if and only if the number of cycles of even length in its cycle decomposition is odd.

For example, $\sigma = (1\ 2\ 3\ 4\ 5\ 6)(7\ 8\ 9)(10\ 11)(12\ 13\ 14\ 15)(16\ 17\ 18)$ has 3 cycles of even length, so $\epsilon(\sigma) = -1$. On the other hand, $\tau = (1\ 12\ 8\ 10\ 4)(2\ 13)(5\ 11\ 7)(6\ 9)$ has exactly 2 cycles of even length, hence $\epsilon(\tau) = 1$.

Be careful not to confuse the terms “odd” and “even” for a permutation σ with the parity of the order of σ . In fact, if σ is of odd order, all cycles in the cycle decomposition of σ have odd length so σ has an even (in this case 0) number of cycles of even length and hence is an even permutation. If $|\sigma|$ is even, σ may be either an even or an odd permutation; e.g., $(1\ 2)$ is odd, $(1\ 2)(3\ 4)$ is even but both have order 2.

As we mentioned in the preceding section, the alternating groups A_n will be important in the study of solvability of polynomials. In the next chapter we shall prove:

A_n is a non-abelian simple group for all $n \geq 5$.

For small values of n , A_n is already familiar to us: A_1 and A_2 are both the trivial group and $|A_3| = 3$ (so $A_3 = \langle (1\ 2\ 3) \rangle \cong Z_3$). The group A_4 has order 12. Exercise 7 shows A_4 is isomorphic to the group of symmetries of a regular tetrahedron. The lattice of subgroups of A_4 appears in Figure 8 (Exercise 8 asserts that this is its complete lattice of subgroups). One of the nicer aspects of this lattice is that (unlike “virtually all groups”) it is a planar graph (there are no crossing lines except at the vertices; see the lattice of D_{16} for a nonplanar lattice).

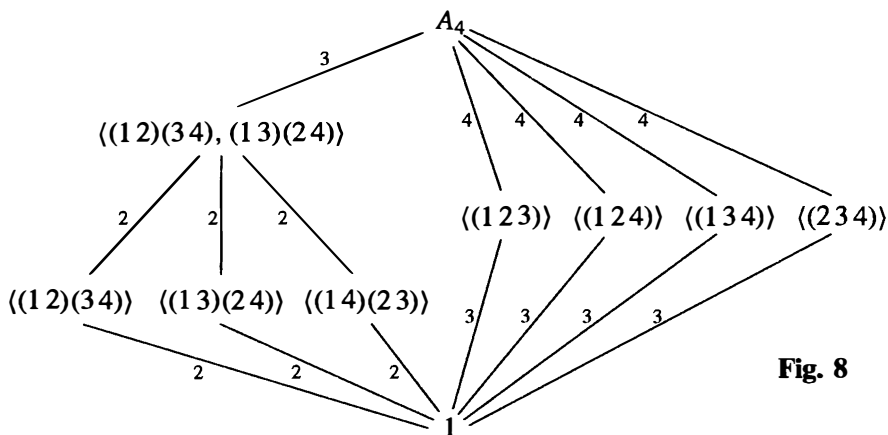


Fig. 8

EXERCISES

- In Exercises 1 and 2 of Section 1.3 you were asked to find the cycle decomposition of some permutations. Write each of these permutations as a product of transpositions. Determine which of these is an even permutation and which is an odd permutation.
- Prove that σ^2 is an even permutation for every permutation σ .
- Prove that S_n is generated by $\{(i\ i+1) \mid 1 \leq i \leq n-1\}$. [Consider conjugates, viz. $(2\ 3)(1\ 2)(2\ 3)^{-1}$.]
- Show that $S_n = \langle (1\ 2), (1\ 2\ 3 \dots n) \rangle$ for all $n \geq 2$.
- Show that if p is prime, $S_p = \langle \sigma, \tau \rangle$ where σ is any transposition and τ is any p -cycle.
- Show that $\langle (1\ 3), (1\ 2\ 3\ 4) \rangle$ is a proper subgroup of S_4 . What is the isomorphism type of this subgroup?
- Prove that the group of rigid motions of a tetrahedron is isomorphic to A_4 . [Recall Exercise 20 in Section 1.7.]
- Prove the lattice of subgroups of A_4 given in the text is correct. [By the preceding exercise and the comments following Lagrange's Theorem, A_4 has no subgroup of order 6.]
- Prove that the (unique) subgroup of order 4 in A_4 is normal and is isomorphic to V_4 .
- Find a composition series for A_4 . Deduce that A_4 is solvable.
- Prove that S_4 has no subgroup isomorphic to Q_8 .
- Prove that A_n contains a subgroup isomorphic to S_{n-2} for each $n \geq 3$.
- Prove that every element of order 2 in A_n is the square of an element of order 4 in S_n . [An element of order 2 in A_n is a product of $2k$ commuting transpositions.]
- Prove that the subgroup of A_4 generated by any element of order 2 and any element of order 3 is all of A_4 .
- Prove that if x and y are distinct 3-cycles in S_4 with $x \neq y^{-1}$, then the subgroup of S_4 generated by x and y is A_4 .
- Let x and y be distinct 3-cycles in S_5 with $x \neq y^{-1}$.
 - Prove that if x and y fix a common element of $\{1, \dots, 5\}$, then $\langle x, y \rangle \cong A_4$.
 - Prove that if x and y do not fix a common element of $\{1, \dots, 5\}$, then $\langle x, y \rangle = A_5$.
- If x and y are 3-cycles in S_n , prove that $\langle x, y \rangle$ is isomorphic to Z_3 , A_4 , A_5 or $Z_3 \times Z_3$.

Group Actions

In this chapter we consider some of the consequences of a group acting on a set. It is an important and recurring idea in mathematics that when one object acts on another then much information can be obtained on both. As more structure is added to the set on which the group acts (for example, groups acting on groups or groups acting on vector spaces (considered in Chapter 18)), more information on the structure of the group becomes available. This study of group actions culminates here in the proof of Sylow's Theorem and the examples and classifications which accrue from it.

The concept of an action will recur as we study modules, vector spaces, canonical forms for matrices and Galois Theory, and is one of the fundamental unifying themes in the text.

4.1 GROUP ACTIONS AND PERMUTATION REPRESENTATIONS

In this section we give the basic theory of group actions and then apply this theory to subgroups of S_n acting on $\{1, 2, \dots, n\}$ to prove that every element of S_n has a unique cycle decomposition. In Sections 2 and 3 we apply the general theory to two other specific group actions to derive some important results.

Let G be a group acting on a nonempty set A . Recall from Section 1.7 that for each $g \in G$ the map

$$\sigma_g : A \rightarrow A \quad \text{defined by} \quad \sigma_g : a \mapsto g \cdot a$$

is a permutation of A . We also saw in Section 1.7 that there is a homomorphism associated to an action of G on A :

$$\varphi : G \rightarrow S_A \quad \text{defined by} \quad \varphi(g) = \sigma_g,$$

called the *permutation representation* associated to the given action. Recall some additional terminology associated to group actions introduced in Sections 1.7 and 2.2.

Definition.

- (1) The *kernel* of the action is the set of elements of G that act trivially on every element of A : $\{g \in G \mid g \cdot a = a \text{ for all } a \in A\}$.
- (2) For each $a \in A$ the *stabilizer* of a in G is the set of elements of G that fix the element a : $\{g \in G \mid g \cdot a = a\}$ and is denoted by G_a .
- (3) An action is *faithful* if its kernel is the identity.

Note that the kernel of an action is precisely the same as the kernel of the associated permutation representation; in particular, the kernel is a normal subgroup of G . Two group elements induce the same permutation on A if and only if they are in the same coset of the kernel (if and only if they are in the same fiber of the permutation representation φ). In particular an action of G on A may also be viewed as a faithful action of the quotient group $G/\ker \varphi$ on A . Recall from Section 2.2 that the stabilizer in G of an element a of A is a subgroup of G . If a is a fixed element of A , then the kernel of the action is contained in the stabilizer G_a since the kernel of the action is the set of elements of G that stabilize every point, namely $\bigcap_{a \in A} G_a$.

Examples

- (1) Let n be a positive integer. The group $G = S_n$ acts on the set $A = \{1, 2, \dots, n\}$ by $\sigma \cdot i = \sigma(i)$ for all $i \in \{1, \dots, n\}$. The permutation representation associated to this action is the identity map $\varphi : S_n \rightarrow S_n$. This action is faithful and for each $i \in \{1, \dots, n\}$ the stabilizer G_i (the subgroup of all permutations fixing i) is isomorphic to S_{n-1} (cf. Exercise 15, Section 3.2).
- (2) Let $G = D_8$ act on the set A consisting of the four vertices of a square. Label these vertices 1,2,3,4 in a clockwise fashion as in Figure 2 of Section 1.2. Let r be the rotation of the square clockwise by $\pi/2$ radians and let s be the reflection in the line which passes through vertices 1 and 3. Then the permutations of the vertices given by r and s are

$$\sigma_r = (1\ 2\ 3\ 4) \quad \text{and} \quad \sigma_s = (2\ 4).$$

Note that since the permutation representation is a homomorphism, the permutation of the four vertices corresponding to sr is $\sigma_{sr} = \sigma_s \sigma_r = (1\ 4)(2\ 3)$. The action of D_8 on the four vertices of a square is faithful since only the identity symmetry fixes all four vertices. The stabilizer of any vertex a is the subgroup of D_8 of order 2 generated by the reflection about the line passing through a and the center of the square (so, for example, the stabilizer of vertex 1 is $\langle s \rangle$).

- (3) Label the four vertices of a square as in the preceding example and now let A be the set whose elements consist of unordered pairs of opposite vertices: $A = \{ \{1, 3\}, \{2, 4\} \}$. Then D_8 also acts on this set A since each symmetry of the square sends a pair of opposite vertices to a pair of opposite vertices. The rotation r interchanges the pairs $\{1, 3\}$ and $\{2, 4\}$; the reflection s fixes both unordered pairs of opposite vertices. Thus if we label the pairs $\{1, 3\}$ and $\{2, 4\}$ as **1** and **2**, respectively, then the permutations of A given by r and s are

$$\sigma_r = (\mathbf{1}\ \mathbf{2}) \quad \text{and} \quad \sigma_s = \text{the identity permutation.}$$

This action of D_8 is not faithful: its kernel is $\langle s, r^2 \rangle$. Moreover, for each $a \in A$ the stabilizer in D_8 of a is the same as the kernel of the action.

- (4) Label the four vertices of a square as in Example 2 and now let A be the following set of unordered pairs of vertices: $\{ \{1, 2\}, \{3, 4\} \}$. The group D_8 does *not* act on this set A because $\{1, 2\} \in A$ but $r \cdot \{1, 2\} = \{2, 3\} \notin A$.

The relation between actions and homomorphisms into symmetric groups may be reversed. Namely, given any nonempty set A and any homomorphism φ of the group G into S_A we obtain an action of G on A by defining

$$g \cdot a = \varphi(g)(a)$$

for all $g \in G$ and all $a \in A$. The kernel of this action is the same as $\ker \varphi$. The permutation representation associated to this action is precisely the given homomorphism φ . This proves the following result.

Proposition 1. For any group G and any nonempty set A there is a bijection between the actions of G on A and the homomorphisms of G into S_A .

In view of Proposition 1 the definition of a permutation representation may be rephrased.

Definition. If G is a group, a *permutation representation* of G is any homomorphism of G into the symmetric group S_A for some nonempty set A . We shall say a given action of G on A *affords* or *induces* the associated permutation representation of G .

We can think of a permutation representation as an analogue of the matrix representation of a linear transformation. In the case where A is a finite set of n elements we have $S_A \cong S_n$ (cf. Section 1.6), so by fixing a labelling of the elements of A we may consider our permutations as elements of the group S_n (which is exactly what we did in Examples 2 and 3 above), in the same way that fixing a basis for a vector space allows us to view a linear transformation as a matrix.

We now prove a combinatorial result about group actions which will have important consequences when we apply it to specific actions in subsequent sections.

Proposition 2. Let G be a group acting on the nonempty set A . The relation on A defined by

$$a \sim b \quad \text{if and only if} \quad a = g \cdot b \text{ for some } g \in G$$

is an equivalence relation. For each $a \in A$, the number of elements in the equivalence class containing a is $|G : G_a|$, the index of the stabilizer of a .

Proof: We first prove $\sim$ is an equivalence relation. By axiom 2 of an action, $a = 1 \cdot a$ for all $a \in A$, i.e., $a \sim a$ and the relation is reflexive. If $a \sim b$, then $a = g \cdot b$ for some $b \in G$ so that

$$g^{-1} \cdot a = g^{-1} \cdot (g \cdot b) = (g^{-1}g) \cdot b = 1 \cdot b = b$$

that is, $b \sim a$ and the relation is symmetric. Finally, if $a \sim b$ and $b \sim c$, then $a = g \cdot b$ and $b = h \cdot c$, for some $g, h \in G$ so

$$a = g \cdot b = g \cdot (h \cdot c) = (gh) \cdot c$$

hence $a \sim c$, and the relation is transitive.

To prove the last statement of the proposition we exhibit a bijection between the left cosets of G_a in G and the elements of the equivalence class of a . Let $\mathcal{C}_a$ be the class of a , so

$$\mathcal{C}_a = \{g \cdot a \mid g \in G\}.$$

Suppose $b = g \cdot a \in \mathcal{C}_a$. Then gG_a is a left coset of G_a in G . The map

$$b = g \cdot a \mapsto gG_a$$

is a map from C_a to the set of left cosets of G_a in G . This map is surjective since for any $g \in G$ the element $g \cdot a$ is an element of C_a . Since $g \cdot a = h \cdot a$ if and only if $h^{-1}g \in G_a$ if and only if $gG_a = hG_a$, the map is also injective, hence is a bijection. This completes the proof.

By Proposition 2 a group G acting on the set A partitions A into disjoint equivalence classes under the action of G . These classes are given a name:

Definition. Let G be a group acting on the nonempty set A .

- (1) The equivalence class $\{g \cdot a \mid g \in G\}$ is called the *orbit* of G containing a .
- (2) The action of G on A is called *transitive* if there is only one orbit, i.e., given any two elements $a, b \in A$ there is some $g \in G$ such that $a = g \cdot b$.

Examples

Let G be a group acting on the set A .

- (1) If G acts trivially on A then $G_a = G$ for all $a \in A$ and the orbits are the elements of A . This action is transitive if and only if $|A| = 1$.
- (2) The symmetric group $G = S_n$ acts transitively in its usual action as permutations on $A = \{1, 2, \dots, n\}$. Note that the stabilizer in G of any point i has index $n = |A|$ in S_n .
- (3) When the group G acts on the set A , any subgroup of G also acts on A . If G is transitive on A a subgroup of G need not be transitive on A . For example, if $G = \langle (1\ 2), (3\ 4) \rangle \leq S_4$ then the orbits of G on $\{1, 2, 3, 4\}$ are $\{1, 2\}$ and $\{3, 4\}$ and there is no element of G that sends 2 to 3. The discussion below on cycle decompositions shows that when $\langle \sigma \rangle$ is any cyclic subgroup of S_n then the orbits of $\langle \sigma \rangle$ consist of the sets of numbers that appear in the individual cycles in the cycle decomposition of σ (for example, the orbits of $\langle (1\ 2)(3\ 4\ 5) \rangle$ are $\{1, 2\}$ and $\{3, 4, 5\}$).
- (4) The group D_8 acts transitively on the four vertices of the square and the stabilizer of any vertex is the subgroup of order 2 (and index 4) generated by the reflection about the line of symmetry passing through that point.
- (5) The group D_8 also acts transitively on the set of two pairs of opposite vertices. In this action the stabilizer of any point is $\langle s, r^2 \rangle$ (which is of index 2).

Cycle Decompositions

We now prove that every element of the symmetric group S_n has the unique cycle decomposition described in Section 1.3. Let $A = \{1, 2, \dots, n\}$, let σ be an element of S_n and let $G = \langle \sigma \rangle$. Then $\langle \sigma \rangle$ acts on A and so, by Proposition 2, it partitions $\{1, 2, \dots, n\}$ into a unique set of (disjoint) orbits. Let $\mathcal{O}$ be one of these orbits and let $x \in \mathcal{O}$. By (the proof of) Proposition 2 applied to $A = \mathcal{O}$ we see that there is a bijection between the left cosets of G_x in G and the elements of $\mathcal{O}$, given explicitly by

$$\sigma^i x \mapsto \sigma^i G_x.$$

Since G is a cyclic group, $G_x \trianglelefteq G$ and G/G_x is cyclic of order d , where d is the smallest positive integer for which $\sigma^d \in G_x$ (cf. Example 2 following Proposition 7 in Section 3.1). Also, $d = |G : G_x| = |\mathcal{O}|$. Thus the distinct cosets of G_x in G are

$$1G_x, \sigma G_x, \sigma^2 G_x, \dots, \sigma^{d-1} G_x.$$

This shows that the distinct elements of $\mathcal{O}$ are

$$x, \sigma(x), \sigma^2(x), \dots, \sigma^{d-1}(x).$$

Ordering the elements of $\mathcal{O}$ in this manner shows that σ cycles the elements of $\mathcal{O}$, that is, on an orbit of size d , σ acts as a d -cycle. This proves the existence of a cycle decomposition for each $\sigma \in S_n$.

The orbits of $\langle \sigma \rangle$ are uniquely determined by σ . The only latitude is in which order the orbits are listed. Within each orbit, $\mathcal{O}$, we may begin with any element as a representative. Choosing $\sigma^i(x)$ instead of x as the initial representative simply produces the elements of $\mathcal{O}$ in the order

$$\sigma^i(x), \sigma^{i+1}(x), \dots, \sigma^{d-1}(x), x, \sigma(x), \dots, \sigma^{i-1}(x),$$

which is a cyclic permutation (forward $i - 1$ terms) of the original list. It follows that the cycle decomposition above is unique up to a rearrangement of the cycles and up to a cyclic permutation of the integers within each cycle.

Subgroups of symmetric groups are called *permutation groups*. For any subgroup G of S_n the orbits of G will refer to its orbits on $\{1, 2, \dots, n\}$. The orbits of an element σ in S_n will mean the orbits of the group $\langle \sigma \rangle$ (namely the sets of integers comprising the cycles in its cycle decomposition).

The exercises below further illustrate how group theoretic information can be obtained from permutation representations.

EXERCISES

Let G be a group and let A be a nonempty set.

1. Let G act on the set A . Prove that if $a, b \in A$ and $b = g \cdot a$ for some $g \in G$, then $G_b = gG_ag^{-1}$ (G_a is the stabilizer of a). Deduce that if G acts transitively on A then the kernel of the action is $\bigcap_{g \in G} gG_ag^{-1}$.
2. Let G be a *permutation group* on the set A (i.e., $G \leq S_A$), let $\sigma \in G$ and let $a \in A$. Prove that $\sigma G_a \sigma^{-1} = G_{\sigma(a)}$. Deduce that if G acts transitively on A then

$$\bigcap_{\sigma \in G} \sigma G_a \sigma^{-1} = 1.$$

3. Assume that G is an abelian, transitive subgroup of S_A . Show that $\sigma(a) \neq a$ for all $\sigma \in G - \{1\}$ and all $a \in A$. Deduce that $|G| = |A|$. [Use the preceding exercise.]
4. Let S_3 act on the set Ω of ordered pairs: $\{(i, j) \mid 1 \leq i, j \leq 3\}$ by $\sigma((i, j)) = (\sigma(i), \sigma(j))$. Find the orbits of S_3 on Ω . For each $\sigma \in S_3$ find the cycle decomposition of σ under this action (i.e., find its cycle decomposition when σ is considered as an element of S_9 — first fix a labelling of these nine ordered pairs). For each orbit $\mathcal{O}$ of S_3 acting on these nine points pick some $a \in \mathcal{O}$ and find the stabilizer of a in S_3 .
5. For each of parts (a) and (b) repeat the preceding exercise but with S_3 acting on the specified set:
 - (a) the set of 27 triples $\{(i, j, k) \mid 1 \leq i, j, k \leq 3\}$
 - (b) the set $\mathcal{P}(\{1, 2, 3\}) - \{\emptyset\}$ of all 7 nonempty subsets of $\{1, 2, 3\}$.
6. As in Exercise 12 of Section 2.2 let R be the set of all polynomials with integer coefficients in the independent variables x_1, x_2, x_3, x_4 and let S_4 act on R by permuting the indices of

the four variables:

$$\sigma \cdot p(x_1, x_2, x_3, x_4) = p(x_{\sigma(1)}, x_{\sigma(2)}, x_{\sigma(3)}, x_{\sigma(4)})$$

for all $\sigma \in S_4$.

- (a) Find the polynomials in the orbit of S_4 on R containing $x_1 + x_2$ (recall from Exercise 12 in Section 2.2 that the stabilizer of this polynomial has order 4).
 - (b) Find the polynomials in the orbit of S_4 on R containing $x_1x_2 + x_3x_4$ (recall from Exercise 12 in Section 2.2 that the stabilizer of this polynomial has order 8).
 - (c) Find the polynomials in the orbit of S_4 on R containing $(x_1 + x_2)(x_3 + x_4)$.
7. Let G be a transitive permutation group on the finite set A . A *block* is a nonempty subset B of A such that for all $\sigma \in G$ either $\sigma(B) = B$ or $\sigma(B) \cap B = \emptyset$ (here $\sigma(B)$ is the set $\{\sigma(b) \mid b \in B\}$).
- (a) Prove that if B is a block containing the element a of A , then the set G_B defined by $G_B = \{\sigma \in G \mid \sigma(B) = B\}$ is a subgroup of G containing G_a .
 - (b) Show that if B is a block and $\sigma_1(B), \sigma_2(B), \dots, \sigma_n(B)$ are all the distinct images of B under the elements of G , then these form a partition of A .
 - (c) A (transitive) group G on a set A is said to be *primitive* if the only blocks in A are the trivial ones: the sets of size 1 and A itself. Show that S_4 is primitive on $A = \{1, 2, 3, 4\}$. Show that D_8 is not primitive as a permutation group on the four vertices of a square.
 - (d) Prove that the transitive group G is primitive on A if and only if for each $a \in A$, the only subgroups of G containing G_a are G_a and G (i.e., G_a is a *maximal* subgroup of G , cf. Exercise 16, Section 2.4). [Use part (a).]
8. A transitive permutation group G on a set A is called *doubly transitive* if for any (hence all) $a \in A$ the subgroup G_a is transitive on the set $A - \{a\}$.
- (a) Prove that S_n is doubly transitive on $\{1, 2, \dots, n\}$ for all $n \geq 2$.
 - (b) Prove that a doubly transitive group is primitive. Deduce that D_8 is not doubly transitive in its action on the 4 vertices of a square.
9. Assume G acts transitively on the finite set A and let H be a normal subgroup of G . Let $\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_r$ be the distinct orbits of H on A .
- (a) Prove that G permutes the sets $\mathcal{O}_1, \mathcal{O}_2, \dots, \mathcal{O}_r$ in the sense that for each $g \in G$ and each $i \in \{1, \dots, r\}$ there is a j such that $g\mathcal{O}_i = \mathcal{O}_j$, where $g\mathcal{O} = \{g \cdot a \mid a \in \mathcal{O}\}$ (i.e., in the notation of Exercise 7 the sets $\mathcal{O}_1, \dots, \mathcal{O}_r$ are blocks). Prove that G is transitive on $\{\mathcal{O}_1, \dots, \mathcal{O}_r\}$. Deduce that all orbits of H on A have the same cardinality.
 - (b) Prove that if $a \in \mathcal{O}_1$ then $|\mathcal{O}_1| = |H : H \cap G_a|$ and prove that $r = |G : HG_a|$. [Draw the sublattice describing the Second Isomorphism Theorem for the subgroups H and G_a of G . Note that $H \cap G_a = H_a$.]
10. Let H and K be subgroups of the group G . For each $x \in G$ define the *HK double coset* of x in G to be the set
- $$HxK = \{h x k \mid h \in H, k \in K\}.$$
- (a) Prove that HxK is the union of the left cosets $x_1K, \dots, x_nK$ where $\{x_1K, \dots, x_nK\}$ is the orbit containing xK of H acting by left multiplication on the set of left cosets of K .
 - (b) Prove that HxK is a union of right cosets of H .
 - (c) Show that HxK and HyK are either the same set or are disjoint for all $x, y \in G$. Show that the set of HK double cosets partitions G .
 - (d) Prove that $|HxK| = |K| \cdot |H : H \cap xKx^{-1}|$.
 - (e) Prove that $|HxK| = |H| \cdot |K : K \cap x^{-1}Hx|$.

4.2 GROUPS ACTING ON THEMSELVES BY LEFT MULTIPLICATION — CAYLEY'S THEOREM

In this section G is any group and we first consider G acting on itself (i.e., $A = G$) by left multiplication:

$$g \cdot a = ga \quad \text{for all } g \in G, a \in G$$

where ga denotes the product of the two group elements g and a in G (if G is written additively, the action will be written $g \cdot a = g + a$ and called left translation). We saw in Section 1.7 that this satisfies the two axioms of a group action.

When G is a finite group of order n it is convenient to label the elements of G with the integers $1, 2, \dots, n$ in order to describe the permutation representation afforded by this action. In this way the elements of G are listed as $g_1, g_2, \dots, g_n$ and for each $g \in G$ the permutation σ_g may be described as a permutation of the indices $1, 2, \dots, n$ as follows:

$$\sigma_g(i) = j \quad \text{if and only if} \quad gg_i = g_j.$$

A different labelling of the group elements will give a different description of σ_g as a permutation of $\{1, 2, \dots, n\}$ (cf. the exercises).

Example

Let $G = \{1, a, b, c\}$ be the Klein 4-group whose group table is written out in Section 2.5. Label the group elements $1, a, b, c$ with the integers $1, 2, 3, 4$, respectively. Under this labelling we compute the permutation σ_a induced by the action of left multiplication by the element a :

$$\begin{aligned} a \cdot 1 &= a1 = a \text{ and so } \sigma_a(1) = 2 \\ a \cdot a &= aa = 1 \text{ and so } \sigma_a(2) = 1 \\ a \cdot b &= ab = c \text{ and so } \sigma_a(3) = 4 \text{ and} \\ a \cdot c &= ac = b \text{ and so } \sigma_a(4) = 3. \end{aligned}$$

With this labelling of the elements of G we see that $\sigma_a = (1\ 2)(3\ 4)$. In the permutation representation associated to the action of the Klein 4-group on itself by left multiplication one similarly computes that

$$a \mapsto \sigma_a = (1\ 2)(3\ 4) \quad b \mapsto \sigma_b = (1\ 3)(2\ 4) \quad c \mapsto \sigma_c = (1\ 4)(2\ 3),$$

which explicitly gives the permutation representation $G \rightarrow S_4$ associated to this action under this labelling.

It is easy to see (and we shall prove this shortly in a more general setting) that the action of a group on itself by left multiplication is always transitive and faithful, and that the stabilizer of any point is the identity subgroup (these facts can be checked by inspection for the above example).

We now consider a generalization of the action of a group by left multiplication on the set of its elements. Let H be any subgroup of G and let A be the set of all left cosets of H in G . Define an action of G on A by

$$g \cdot aH = gaH \quad \text{for all } g \in G, aH \in A$$

where gaH is the left coset with representative ga . One easily checks that this satisfies the two axioms for a group action, i.e., that G does act on the set of left cosets of H

by left multiplication. In the special case when H is the identity subgroup of G the coset aH is just $\{a\}$ and if we identify the element a with the set $\{a\}$, this action by left multiplication on left cosets of the identity subgroup is the same as the action of G on itself by left multiplication.

When H is of finite index m in G it is convenient to label the left cosets of H with the integers $1, 2, \dots, m$ in order to describe the permutation representation afforded by this action. In this way the distinct left cosets of H in G are listed as $a_1H, a_2H, \dots, a_mH$ and for each $g \in G$ the permutation σ_g may be described as a permutation of the indices $1, 2, \dots, m$ as follows:

$$\sigma_g(i) = j \quad \text{if and only if} \quad ga_iH = a_jH.$$

A different labelling of the group elements will give a different description of σ_g as a permutation of $\{1, 2, \dots, m\}$ (cf. the exercises).

Example

Let $G = D_8$ and let $H = \langle s \rangle$. Label the distinct left cosets $1H, rH, r^2H, r^3H$ with the integers 1, 2, 3, 4 respectively. Under this labelling we compute the permutation σ_s induced by the action of left multiplication by the group element s on the left cosets of H :

$$\begin{aligned} s \cdot 1H &= sH = 1H \text{ and so } \sigma_s(1) = 1 \\ s \cdot rH &= srH = r^3H \text{ and so } \sigma_s(2) = 4 \\ s \cdot r^2H &= sr^2H = r^2H \text{ and so } \sigma_s(3) = 3 \\ s \cdot r^3H &= sr^3H = rH \text{ and so } \sigma_s(4) = 2. \end{aligned}$$

With this labelling of the left cosets of H we obtain $\sigma_s = (2 \ 4)$. In the permutation representation associated to the action of D_8 on the left cosets of $\langle s \rangle$ by left multiplication one similarly computes that $\sigma_r = (1 \ 2 \ 3 \ 4)$. Note that the permutation representation is a homomorphism, so once its value has been determined on generators for D_8 its value on any other element can be determined (e.g., $\sigma_{sr^2} = \sigma_s \sigma_r^2$).

Theorem 3. Let G be a group, let H be a subgroup of G and let G act by left multiplication on the set A of left cosets of H in G . Let π_H be the associated permutation representation afforded by this action. Then

- (1) G acts transitively on A
- (2) the stabilizer in G of the point $1H \in A$ is the subgroup H
- (3) the kernel of the action (i.e., the kernel of π_H) is $\bigcap_{x \in G} xHx^{-1}$, and $\ker \pi_H$ is the largest normal subgroup of G contained in H .

Proof: To see that G acts transitively on A , let aH and bH be any two elements of A , and let $g = ba^{-1}$. Then $g \cdot aH = (ba^{-1})aH = bH$, and so the two arbitrary elements aH and bH of A lie in the same orbit, which proves (1). For (2), the stabilizer of the point $1H$ is, by definition, $\{g \in G \mid g \cdot 1H = 1H\}$, i.e., $\{g \in G \mid gH = H\} = H$.

By definition of π_H we have

$$\begin{aligned} \ker \pi_H &= \{g \in G \mid gxH = xH \text{ for all } x \in G\} \\ &= \{g \in G \mid (x^{-1}gx)H = H \text{ for all } x \in G\} \\ &= \{g \in G \mid x^{-1}gx \in H \text{ for all } x \in G\} \\ &= \{g \in G \mid g \in xHx^{-1} \text{ for all } x \in G\} = \bigcap_{x \in G} xHx^{-1}, \end{aligned}$$

which proves the first assertion of (3). The second assertion of (3) comes from observing first that $\ker \pi_H \trianglelefteq G$ and $\ker \pi_H \leq H$. If now N is any normal subgroup of G contained in H then we have $N = xNx^{-1} \leq xHx^{-1}$ for all $x \in G$ so that

$$N \leq \bigcap_{x \in G} xHx^{-1} = \ker \pi_H.$$

This shows that $\ker \pi_H$ is the largest normal subgroup of G contained in H .

Corollary 4. (Cayley's Theorem) Every group is isomorphic to a subgroup of some symmetric group. If G is a group of order n , then G is isomorphic to a subgroup of S_n .

Proof: Let $H = 1$ and apply the preceding theorem to obtain a homomorphism of G into S_G (here we are identifying the cosets of the identity subgroup with the elements of G). Since the kernel of this homomorphism is contained in $H = 1$, G is isomorphic to its image in S_G .

Note that G is isomorphic to a *subgroup* of a symmetric group, not to the full symmetric group itself. For example, we exhibited an isomorphism of the Klein 4-group with the subgroup $\langle (1\ 2)(3\ 4), (1\ 3)(2\ 4) \rangle$ of S_4 . Recall that subgroups of symmetric groups are called *permutation groups* so Cayley's Theorem states that every group is isomorphic to a permutation group. The permutation representation afforded by left multiplication on the elements of G (cosets of $H = 1$) is called the *left regular representation* of G . One might think that we could study all groups more effectively by simply studying subgroups of symmetric groups (and all finite groups by studying subgroups of S_n , for all n). This approach alone is neither computationally nor theoretically practical, since to study groups of order n we would have to work in the much larger group S_n (cf. Exercise 7, for example).

Historically, finite groups were first studied not in an axiomatic setting as we have developed but as subgroups of S_n . Thus Cayley's Theorem proves that the historical notion of a group and the modern (axiomatic) one are equivalent. One advantage of the modern approach is that we are not, in our study of a given group, restricted to considering that group as a subgroup of some *particular* symmetric group (so in some sense our groups are "coordinate free").

The next result generalizes our result on the normality of subgroups of index 2.

Corollary 5. If G is a finite group of order n and p is the smallest prime dividing $|G|$, then any subgroup of index p is normal.

Remark: In general, a group of order n need not have a subgroup of index p (for example, A_4 has no subgroup of index 2).

Proof: Suppose $H \leq G$ and $|G : H| = p$. Let π_H be the permutation representation afforded by multiplication on the set of left cosets of H in G , let $K = \ker \pi_H$ and let $|H : K| = k$. Then $|G : K| = |G : H||H : K| = pk$. Since H has p left cosets, G/K is isomorphic to a subgroup of S_p (namely, the image of G under π_H) by the First Isomorphism Theorem. By Lagrange's Theorem, $pk = |G/K|$ divides $p!$.

Thus $k \mid \frac{p!}{p} = (p-1)!$. But all prime divisors of $(p-1)!$ are less than p and by the minimality of p , every prime divisor of k is greater than or equal to p . This forces $k = 1$, so $H = K \leq G$, completing the proof.

EXERCISES

Let G be a group and let H be a subgroup of G .

- Let $G = \{1, a, b, c\}$ be the Klein 4-group whose group table is written out in Section 2.5.
 - Label $1, a, b, c$ with the integers $1, 2, 4, 3$, respectively, and prove that under the left regular representation of G into S_4 the nonidentity elements are mapped as follows:

$$a \mapsto (1\ 2)(3\ 4) \quad b \mapsto (1\ 4)(2\ 3) \quad c \mapsto (1\ 3)(2\ 4).$$

- Relabel $1, a, b, c$ as $1, 4, 2, 3$, respectively, and compute the image of each element of G under the left regular representation of G into S_4 . Show that the image of G in S_4 under this labelling is the same *subgroup* as the image of G in part (a) (even though the nonidentity elements individually map to different permutations under the two different labellings).
- List the elements of S_3 as $1, (1\ 2), (2\ 3), (1\ 3), (1\ 2\ 3), (1\ 3\ 2)$ and label these with the integers $1, 2, 3, 4, 5, 6$ respectively. Exhibit the image of each element of S_3 under the left regular representation of S_3 into S_6 .
 - Let r and s be the usual generators for the dihedral group of order 8.
 - List the elements of D_8 as $1, r, r^2, r^3, s, sr, sr^2, sr^3$ and label these with the integers $1, 2, \dots, 8$ respectively. Exhibit the image of each element of D_8 under the left regular representation of D_8 into S_8 .
 - Relabel this same list of elements of D_8 with the integers $1, 3, 5, 7, 2, 4, 6, 8$ respectively and recompute the image of each element of D_8 under the left regular representation with respect to this new labelling. Show that the two subgroups of S_8 obtained in parts (a) and (b) are different.
 - Use the left regular representation of Q_8 to produce two elements of S_8 which generate a subgroup of S_8 isomorphic to the quaternion group Q_8 .
 - Let r and s be the usual generators for the dihedral group of order 8 and let $H = \langle s \rangle$. List the left cosets of H in D_8 as $1H, rH, r^2H$ and r^3H .
 - Label these cosets with the integers $1, 2, 3, 4$, respectively. Exhibit the image of each element of D_8 under the representation π_H of D_8 into S_4 obtained from the action of D_8 by left multiplication on the set of 4 left cosets of H in D_8 . Deduce that this representation is faithful (i.e., the elements of S_4 obtained form a subgroup isomorphic to D_8).
 - Repeat part (a) with the list of cosets relabelled by the integers $1, 3, 2, 4$, respectively. Show that the permutations obtained from this labelling form a subgroup of S_4 that is different from the subgroup obtained in part (a).
 - Let $K = \langle sr \rangle$, list the cosets of K in D_8 as $1K, rK, r^2K$ and r^3K , and label these with the integers $1, 2, 3, 4$. Prove that, with respect to this labelling, the image of D_8 under the representation π_K obtained from left multiplication on the cosets of K is the same *subgroup* of S_4 as in part (a) (even though the subgroups H and K are different and some of the elements of D_8 map to different permutations under the two homomorphisms).

6. Let r and s be the usual generators for the dihedral group of order 8 and let $N = \langle r^2 \rangle$. List the left cosets of N in D_8 as $1N$, rN , sN and srN . Label these cosets with the integers 1, 2, 3, 4 respectively. Exhibit the image of each element of D_8 under the representation π_N of D_8 into S_4 obtained from the action of D_8 by left multiplication on the set of 4 left cosets of N in D_8 . Deduce that this representation is not faithful and prove that $\pi_N(D_8)$ is isomorphic to the Klein 4-group.
7. Let Q_8 be the quaternion group of order 8.
 - (a) Prove that Q_8 is isomorphic to a subgroup of S_8 .
 - (b) Prove that Q_8 is not isomorphic to a subgroup of S_n for any $n \leq 7$. [If Q_8 acts on any set A of order ≤ 7 show that the stabilizer of any point $a \in A$ must contain the subgroup $\langle -1 \rangle$.]
8. Prove that if H has finite index n then there is a normal subgroup K of G with $K \leq H$ and $|G : K| \leq n!$.
9. Prove that if p is a prime and G is a group of order p^α for some $\alpha \in \mathbb{Z}^+$, then every subgroup of index p is normal in G . Deduce that every group of order p^2 has a normal subgroup of order p .
10. Prove that every non-abelian group of order 6 has a nonnormal subgroup of order 2. Use this to classify groups of order 6. [Produce an injective homomorphism into S_3 .]
11. Let G be a finite group and let $\pi : G \rightarrow S_G$ be the left regular representation. Prove that if x is an element of G of order n and $|G| = mn$, then $\pi(x)$ is a product of m n -cycles. Deduce that $\pi(x)$ is an odd permutation if and only if $|x|$ is even and $\frac{|G|}{|x|}$ is odd.
12. Let G and π be as in the preceding exercise. Prove that if $\pi(G)$ contains an odd permutation then G has a subgroup of index 2. [Use Exercise 3 in Section 3.3.]
13. Prove that if $|G| = 2k$ where k is odd then G has a subgroup of index 2. [Use Cauchy's Theorem to produce an element of order 2 and then use the preceding two exercises.]
14. Let G be a finite group of composite order n with the property that G has a subgroup of order k for each positive integer k dividing n . Prove that G is not simple.

4.3 GROUPS ACTING ON THEMSELVES BY CONJUGATION —THE CLASS EQUATION

In this section G is any group and we first consider G acting on itself (i.e., $A = G$) by conjugation:

$$g \cdot a = gag^{-1} \quad \text{for all } g \in G, a \in G$$

where gag^{-1} is computed in the group G as usual. This definition satisfies the two axioms for a group action because

$$g_1 \cdot (g_2 \cdot a) = g_1 \cdot (g_2 a g_2^{-1}) = g_1 (g_2 a g_2^{-1}) g_1^{-1} = (g_1 g_2) a (g_1 g_2)^{-1} = (g_1 g_2) \cdot a$$

and

$$1 \cdot a = 1a1^{-1} = a$$

for all $g_1, g_2 \in G$ and all $a \in G$.

Definition. Two elements a and b of G are said to be *conjugate in G* if there is some $g \in G$ such that $b = gag^{-1}$ (i.e., if and only if they are in the same orbit of G acting on itself by conjugation). The orbits of G acting on itself by conjugation are called the *conjugacy classes of G* .

Examples

- (1) If G is an abelian group then the action of G on itself by conjugation is the trivial action: $g \cdot a = a$, for all $g, a \in G$, and for each $a \in G$ the conjugacy class of a is $\{a\}$.
- (2) If $|G| > 1$ then, unlike the action by left multiplication, G does *not* act transitively on itself by conjugation because $\{1\}$ is always a conjugacy class (i.e., an orbit for this action). More generally, the one element subset $\{a\}$ is a conjugacy class if and only if $gag^{-1} = a$ for all $g \in G$ if and only if a is in the center of G .
- (3) In S_3 one can compute directly that the conjugacy classes are $\{1\}$, $\{(1\ 2), (1\ 3), (2\ 3)\}$ and $\{(1\ 2\ 3), (1\ 3\ 2)\}$. We shall shortly develop techniques for computing conjugacy classes more easily, particularly in symmetric groups.

As in the case of a group acting on itself by left multiplication, the action by conjugation can be generalized. If S is any subset of G , define

$$gSg^{-1} = \{gs g^{-1} \mid s \in S\}.$$

A group G acts on the set $\mathcal{P}(G)$ of all subsets of itself by defining $g \cdot S = gSg^{-1}$ for any $g \in G$ and $S \in \mathcal{P}(G)$. As above, this defines a group action of G on $\mathcal{P}(G)$. Note that if S is the one element set $\{s\}$ then $g \cdot S$ is the one element set $\{gs g^{-1}\}$ and so this action of G on all subsets of G may be considered as an extension of the action of G on itself by conjugation.

Definition. Two subsets S and T of G are said to be *conjugate in G* if there is some $g \in G$ such that $T = gSg^{-1}$ (i.e., if and only if they are in the same orbit of G acting on its subsets by conjugation).

We now apply Proposition 2 to the action of G by conjugation. Proposition 2 proves that if S is a subset of G , then the number of conjugates of S equals the index $|G : G_S|$ of the stabilizer G_S of S . For action by conjugation

$$G_S = \{g \in G \mid gSg^{-1} = S\} = N_G(S)$$

is the normalizer of S in G . We summarize this as

Proposition 6. The number of conjugates of a subset S in a group G is the index of the normalizer of S , $|G : N_G(S)|$. In particular, the number of conjugates of an element s of G is the index of the centralizer of s , $|G : C_G(s)|$.

Proof: The second assertion of the proposition follows from the observation that $N_G(\{s\}) = C_G(s)$.

The action of G on itself by conjugation partitions G into the conjugacy classes of G , whose orders can be computed by Proposition 6. Since the sum of the orders of these conjugacy classes is the order of G , we obtain the following important relation among these orders.

Theorem 7. (The Class Equation) Let G be a finite group and let $g_1, g_2, \dots, g_r$ be representatives of the distinct conjugacy classes of G not contained in the center $Z(G)$ of G . Then

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|.$$

Proof: As noted in Example 2 above the element $\{x\}$ is a conjugacy class of size 1 if and only if $x \in Z(G)$, since then $gxg^{-1} = x$ for all $g \in G$. Let $Z(G) = \{1, z_2, \dots, z_m\}$, let $\mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_r$ be the conjugacy classes of G not contained in the center, and let g_i be a representative of $\mathcal{K}_i$ for each i . Then the full set of conjugacy classes of G is given by

$$\{1\}, \{z_2\}, \dots, \{z_m\}, \mathcal{K}_1, \mathcal{K}_2, \dots, \mathcal{K}_r.$$

Since these partition G we have

$$\begin{aligned} |G| &= \sum_{i=1}^m 1 + \sum_{i=1}^r |\mathcal{K}_i| \\ &= |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|, \end{aligned}$$

where $|\mathcal{K}_i|$ is given by Proposition 6. This proves the class equation.

Note in particular that all the summands on the right hand side of the class equation are divisors of the group order since they are indices of subgroups of G . This restricts their possible values (cf. Exercise 6, for example).

Examples

- (1) The class equation gives no information in an abelian group since conjugation is the trivial action and all conjugacy classes have size 1.
- (2) In any group G we have $\langle g \rangle \leq C_G(g)$; this observation helps to minimize computations of conjugacy classes. For example, in the quaternion group Q_8 we see that $\langle i \rangle \leq C_{Q_8}(i) \leq Q_8$. Since $i \notin Z(Q_8)$ and $|Q_8 : \langle i \rangle| = 2$, we must have $C_{Q_8}(i) = \langle i \rangle$. Thus i has precisely 2 conjugates in Q_8 , namely i and $-i = kik^{-1}$. The other conjugacy classes in Q_8 are determined similarly and are

$$\{1\}, \{-1\}, \{\pm i\}, \{\pm j\}, \{\pm k\}.$$

The first two classes form $Z(Q_8)$ and the class equation for this group is

$$|Q_8| = 2 + 2 + 2 + 2.$$

- (3) In D_8 we may also use the fact that the three subgroups of index 2 are abelian to quickly see that if $x \notin Z(D_8)$, then $|C_{D_8}(x)| = 4$. The conjugacy classes of D_8 are

$$\{1\}, \{r^2\}, \{r, r^3\}, \{s, sr^2\}, \{sr, sr^3\}.$$

The first two classes form $Z(D_8)$ and the class equation for this group is

$$|D_8| = 2 + 2 + 2 + 2.$$

Before discussing more examples of conjugacy we give two important consequences of the class equation. The first application of the class equation is to show that groups of prime power order have nontrivial centers, which is the starting point for the study of groups of prime power order (to which we return in Chapter 6).

Theorem 8. If p is a prime and P is a group of prime power order p^α for some $\alpha \geq 1$, then P has a nontrivial center: $Z(P) \neq 1$.

Proof: By the class equation

$$|P| = |Z(P)| + \sum_{i=1}^r |P : C_P(g_i)|$$

where $g_1, \dots, g_r$ are representatives of the distinct non-central conjugacy classes. By definition, $C_P(g_i) \neq P$ for $i = 1, 2, \dots, r$ so p divides $|P : C_P(g_i)|$. Since p also divides $|P|$ it follows that p divides $|Z(P)|$, hence the center must be nontrivial.

Corollary 9. If $|P| = p^2$ for some prime p , then P is abelian. More precisely, P is isomorphic to either Z_{p^2} or $Z_p \times Z_p$.

Proof: Since $Z(P) \neq 1$ by the theorem, it follows that $P/Z(P)$ is cyclic. By Exercise 36, Section 3.1, P is abelian. If P has an element of order p^2 , then P is cyclic. Assume therefore that every nonidentity element of P has order p . Let x be any nonidentity element of P and let $y \in P - \langle x \rangle$. Since $|\langle x, y \rangle| > |\langle x \rangle| = p$, we must have that $P = \langle x, y \rangle$. Both x and y have order p so $\langle x \rangle \times \langle y \rangle = Z_p \times Z_p$. It now follows directly that the map $(x^a, y^b) \mapsto x^a y^b$ is an isomorphism from $\langle x \rangle \times \langle y \rangle$ onto P . This completes the proof.

Conjugacy in S_n

We next consider conjugation in symmetric groups. Readers familiar with linear algebra will recognize that in the matrix group $GL_n(F)$, conjugation is the same as “change of basis”: $A \mapsto PAP^{-1}$. The situation in S_n is analogous:

Proposition 10. Let σ, τ be elements of the symmetric group S_n and suppose σ has cycle decomposition

$$(a_1 a_2 \dots a_{k_1}) (b_1 b_2 \dots b_{k_2}) \dots$$

Then $\tau\sigma\tau^{-1}$ has cycle decomposition

$$(\tau(a_1) \tau(a_2) \dots \tau(a_{k_1})) (\tau(b_1) \tau(b_2) \dots \tau(b_{k_2})) \dots,$$

that is, $\tau\sigma\tau^{-1}$ is obtained from σ by replacing each entry i in the cycle decomposition for σ by the entry $\tau(i)$.

Proof: Observe that if $\sigma(i) = j$, then

$$\tau\sigma\tau^{-1}(\tau(i)) = \tau(j).$$

Thus, if the ordered pair i, j appears in the cycle decomposition of σ , then the ordered pair $\tau(i), \tau(j)$ appears in the cycle decomposition of $\tau\sigma\tau^{-1}$. This completes the proof.

Example

Let $\sigma = (1\ 2)(3\ 4\ 5)(6\ 7\ 8\ 9)$ and let $\tau = (1\ 3\ 5\ 7)(2\ 4\ 6\ 8)$. Then

$$\tau\sigma\tau^{-1} = (3\ 4)(5\ 6\ 7)(8\ 1\ 2\ 9).$$

Definition.

- (1) If $\sigma \in S_n$ is the product of disjoint cycles of lengths $n_1, n_2, \dots, n_r$ with $n_1 \leq n_2 \leq \dots \leq n_r$ (including its 1-cycles) then the integers $n_1, n_2, \dots, n_r$ are called the *cycle type* of σ .
- (2) If $n \in \mathbb{Z}^+$, a *partition* of n is any nondecreasing sequence of positive integers whose sum is n .

Note that by the results of the preceding section the cycle type of a permutation is unique. For example, the cycle type of an m -cycle in S_n is $1, 1, \dots, 1, m$, where the m is preceded by $n - m$ ones.

Proposition 11. Two elements of S_n are conjugate in S_n if and only if they have the same cycle type. The number of conjugacy classes of S_n equals the number of partitions of n .

Proof: By Proposition 10, conjugate permutations have the same cycle type. Conversely, suppose the permutations σ_1 and σ_2 have the same cycle type. Order the cycles in nondecreasing length, including 1-cycles (if several cycles of σ_1 and σ_2 have the same length then there are several ways of doing this). Ignoring parentheses, each cycle decomposition is a list in which all the integers from 1 to n appear exactly once. Define τ to be the function which maps the i^{th} integer in the list for σ_1 to the i^{th} integer in the list for σ_2 . Thus τ is a permutation and since the parentheses which delineate the cycle decompositions appear at the same positions in each list, Proposition 10 ensures that $\tau\sigma_1\tau^{-1} = \sigma_2$, so that σ_1 and σ_2 are conjugate.

Since there is a bijection between the conjugacy classes of S_n and the permissible cycle types and each cycle type for a permutation in S_n is a partition of n , the second assertion of the proposition follows, completing the proof.

Examples

- (1) Let $\sigma_1 = (1)(3\ 5)(8\ 9)(2\ 4\ 7\ 6)$ and let $\sigma_2 = (3)(4\ 7)(8\ 1)(5\ 2\ 6\ 9)$. Then define τ by $\tau(1) = 3, \tau(3) = 4, \tau(5) = 7, \tau(8) = 8$, etc. Then

$$\tau = (1\ 3\ 4\ 2\ 5\ 7\ 6\ 9)(8)$$

and $\tau\sigma_1\tau^{-1} = \sigma_2$.

- (2) If in the previous example we had reordered σ_2 as $\sigma_2 = (3)(8\ 1)(4\ 7)(5\ 2\ 6\ 9)$ by interchanging the two cycles of length 2, then the corresponding τ described above is defined by $\tau(1) = 3, \tau(3) = 8, \tau(5) = 1, \tau(8) = 4$, etc., which gives the permutation

$$\tau = (1\ 3\ 8\ 4\ 2\ 5)(6\ 9\ 7)$$

again with $\tau\sigma_1\tau^{-1} = \sigma_2$, which shows that there are many elements conjugating σ_1 into σ_2 .

- (3) If $n = 5$, the partitions of 5 and corresponding representatives of the conjugacy classes (with 1-cycles not written) are as given in the following table:

Partition of 5	Representative of Conjugacy Class
1, 1, 1, 1, 1	1
1, 1, 1, 2	(1 2)
1, 1, 3	(1 2 3)
1, 4	(1 2 3 4)
5	(1 2 3 4 5)
1, 2, 2	(1 2)(3 4)
2, 3	(1 2)(3 4 5)

Proposition 11 and Proposition 6 can be used to exhibit the centralizers of some elements in S_n . For example, if σ is an m -cycle in S_n , then the number of conjugates of σ (i.e., the number of m -cycles) is

$$\frac{n \cdot (n-1) \cdots (n-m+1)}{m}.$$

By Proposition 6 this is the index of the centralizer of σ : $\frac{|S_n|}{|C_{S_n}(\sigma)|}$. Since $|S_n| = n!$ we obtain

$$|C_{S_n}(\sigma)| = m \cdot (n-m)!.$$

The element σ certainly commutes with $1, \sigma, \sigma^2, \dots, \sigma^{m-1}$. It also commutes with any permutation in S_n whose cycles are disjoint from σ and there are $(n-m)!$ permutations of this type (the full symmetric group on the numbers not appearing in σ). The product of elements of these two types already accounts for $m \cdot (n-m)!$ elements commuting with σ . By the order computation above, this is the full centralizer of σ in S_n . Explicitly,

$$\text{if } \sigma \text{ is an } m\text{-cycle in } S_n, \text{ then } C_{S_n}(\sigma) = \{\sigma^i \tau \mid 0 \leq i \leq m-1, \tau \in S_{n-m}\}$$

where S_{n-m} denotes the subgroup of S_n which fixes all integers appearing in the m -cycle σ (and is the identity subgroup if $m = n$ or $m = n-1$).

For example, the centralizer of $\sigma = (1\ 3\ 5)$ in S_7 is the subgroup

$$\{(1\ 3\ 5)^i \tau \mid i = 0, 1 \text{ or } 2, \text{ and } \tau \text{ fixes } 1, 3 \text{ and } 5\}.$$

Note that $\tau \in S_A$ where $A = \{2, 4, 6, 7\}$, so there are $4!$ choices for τ and the centralizer has order $3 \cdot 4! = 72$.

We shall discuss centralizers of other elements of S_n in the next exercises and in Chapter 5.

We can use this discussion of the conjugacy classes in S_n to give a combinatorial proof of the simplicity of A_5 . We first observe that normal subgroups of a group G are the union of conjugacy classes of G , i.e.,

$$\text{if } H \trianglelefteq G, \text{ then for every conjugacy class } \mathcal{K} \text{ of } G \text{ either } \mathcal{K} \subseteq H \text{ or } \mathcal{K} \cap H = \emptyset.$$

This is because if $x \in \mathcal{K} \cap H$, then $gxg^{-1} \in gHg^{-1}$ for all $g \in G$. Since H is normal, $gHg^{-1} = H$, so that H contains all the conjugates of x , i.e., $\mathcal{K} \subseteq H$.

Theorem 12. A_5 is a simple group.

Proof: We first work out the conjugacy classes of A_5 and their orders. Proposition 11 does not apply directly since two elements of the same cycle type (which are conjugate in S_5) need *not* be conjugate in A_5 . Exercises 19 to 22 analyze the relation of classes in S_n to classes in A_n in detail.

We have already seen that representatives of the cycle types of even permutations can be taken to be

$$1, \quad (1\ 2\ 3), \quad (1\ 2\ 3\ 4\ 5) \quad \text{and} \quad (1\ 2)(3\ 4).$$

The centralizers of 3-cycles and 5-cycles in S_5 were determined above, and checking which of these elements are contained in A_5 we see that

$$C_{A_5}((1\ 2\ 3)) = \langle (1\ 2\ 3) \rangle \quad \text{and} \quad C_{A_5}((1\ 2\ 3\ 4\ 5)) = \langle (1\ 2\ 3\ 4\ 5) \rangle.$$

These groups have orders 3 and 5 (index 20 and 12), respectively, so there are 20 distinct conjugates of $(1\ 2\ 3)$ and 12 distinct conjugates of $(1\ 2\ 3\ 4\ 5)$ in A_5 . Since there are a total of twenty 3-cycles in S_5 (Exercise 16, Section 1.3) and all of these lie in A_5 , we see that

all twenty 3-cycles are conjugate in A_5 .

There are a total of twenty-four 5-cycles in A_5 but only 12 distinct conjugates of the 5-cycle $(1\ 2\ 3\ 4\ 5)$. Thus some 5-cycle, σ , is *not* conjugate to $(1\ 2\ 3\ 4\ 5)$ in A_5 (in fact, $(1\ 3\ 5\ 2\ 4)$ is not conjugate in A_5 to $(1\ 2\ 3\ 4\ 5)$ since the method of proof in Proposition 11 shows that any element of S_5 conjugating $(1\ 2\ 3\ 4\ 5)$ into $(1\ 3\ 5\ 2\ 4)$ must be an odd permutation). As above we see that σ also has 12 distinct conjugates in A_5 , hence

the 5-cycles lie in two conjugacy classes in A_5 , each of which has 12 elements.

Since the 3-cycles and 5-cycles account for all the nonidentity elements of odd order, the 15 remaining nonidentity elements of A_5 must have order 2 and therefore have cycle type $(2,2)$. It is easy to see that $(1\ 2)(3\ 4)$ commutes with $(1\ 3)(2\ 4)$ but does not commute with any element of odd order in A_5 . It follows that $|C_{A_5}((12)(34))| = 4$. Thus $(1\ 2)(3\ 4)$ has 15 distinct conjugates in A_5 , hence

all 15 elements of order 2 in A_5 are conjugate to $(1\ 2)(3\ 4)$.

In summary, the conjugacy classes of A_5 have orders 1, 15, 20, 12 and 12.

Now, suppose H were a normal subgroup of A_5 . Then as we observed above, H would be the union of conjugacy classes of A_5 . Then the order of H would be both a divisor of 60 (the order of A_5) and be the sum of some collection of the integers $\{1, 12, 12, 15, 20\}$ (the sizes of the conjugacy classes in A_5). A quick check shows the only possibilities are $|H| = 1$ or $|H| = 60$, so that A_5 has no proper, nontrivial normal subgroups.

Right Group Actions

As noted in Section 1.7, in the definition of an action the group elements appear to the left of the set elements and so our notion of an action might more precisely be termed a *left group action*. One can analogously define the notion of a *right group action* of the

group G on the nonempty set A as a map from $A \times G$ to A , denoted by $a \cdot g$ for $a \in A$ and $g \in G$, that satisfies the axioms:

- (1) $(a \cdot g_1) \cdot g_2 = a \cdot (g_1 g_2)$ for all $a \in A$, and $g_1, g_2 \in G$, and
- (2) $a \cdot 1 = a$ for all $a \in A$.

In much of the literature on group theory, conjugation is written as a right group action using the following notation:

$$a^g = g^{-1}ag \quad \text{for all } g, a \in G.$$

Similarly, for subsets S of G one defines $S^g = g^{-1}Sg$. In this notation the two axioms for a right action are verified as follows:

$$(a^{g_1})^{g_2} = g_2^{-1}(g_1^{-1}ag_1)g_2 = (g_1 g_2)^{-1}a(g_1 g_2) = a^{(g_1 g_2)}$$

and

$$a^1 = 1^{-1}a1 = a$$

for all $g_1, g_2, a \in G$. Thus the two axioms for this right action of a group on itself take the form of the familiar “laws of exponentiation.” (Note that the integer power a^n of a group element a is easily distinguished from the conjugate a^g of a by the nature of the exponent: $n \in \mathbb{Z}$ but $g \in G$.) Because conjugation is so ubiquitous in the theory of groups, this notation is a useful and efficient shorthand (as opposed to always writing gag^{-1} or $g \cdot a$ for action on the left by conjugation).

For arbitrary group actions it is an easy exercise to check that if we are given a left group action of G on A then the map $A \times G \rightarrow A$ defined by $a \cdot g = g^{-1} \cdot a$ is a right group action. Conversely, given a right group action of G on A we can form a left group action by $g \cdot a = a \cdot g^{-1}$. Call these pairs *corresponding group actions*. Put another way, for corresponding group actions, g acts on the left in the same way that g^{-1} acts on the right. This is particularly transparent for the action of conjugation because the “left conjugate of a by g ,” namely gag^{-1} , is the same group element as the “right conjugate of a by g^{-1} ,” namely $a^{g^{-1}}$. Thus two elements or subsets of a group are “left conjugate” if and only if they are “right conjugate,” and so the relation “conjugacy” is the same for the left and right corresponding actions. More generally, it is also an exercise (Exercise 1) to see that for any corresponding left and right actions the orbits are the same.

We have consistently used left actions since they are compatible with the notation of applying functions on the left (i.e., with the notation $\varphi(g)$); in this way left multiplication on the left cosets of a subgroup is a left action. Similarly, right multiplication on the right cosets of a subgroup is a right action and the associated permutation representation φ is a homomorphism provided the function $\varphi : G \rightarrow S_A$ is written on the right as $(g_1 g_2)\varphi$ (and also provided permutations in S_A are written on the right as functions from A to itself). There are instances where a set admits two actions by a group G : one naturally on the left and the other on the right, so that it is useful to be comfortable with both types of actions.

EXERCISES

Let G be a group.

- Suppose G has a left action on a set A , denoted by $g \cdot a$ for all $g \in G$ and $a \in A$. Denote the corresponding right action on A by $a \cdot g$. Prove that the (equivalence) relations $\sim$ and $\sim'$ defined by

$$a \sim b \quad \text{if and only if} \quad a = g \cdot b \quad \text{for some } g \in G$$

and

$$a \sim' b \quad \text{if and only if} \quad a = b \cdot g \quad \text{for some } g \in G$$

are the same relation (i.e., $a \sim b$ if and only if $a \sim' b$).

- Find all conjugacy classes and their sizes in the following groups:
(a) D_8 (b) Q_8 (c) A_4 .
- Find all the conjugacy classes and their sizes in the following groups:
(a) $Z_2 \times S_3$ (b) $S_3 \times S_3$ (c) $Z_3 \times A_4$.
- Prove that if $S \subseteq G$ and $g \in G$ then $gN_G(S)g^{-1} = N_G(gSg^{-1})$ and $gC_G(S)g^{-1} = C_G(gSg^{-1})$.
- If the center of G is of index n , prove that every conjugacy class has at most n elements.
- Assume G is a non-abelian group of order 15. Prove that $Z(G) = 1$. Use the fact that $\langle g \rangle \leq C_G(g)$ for all $g \in G$ to show that there is at most one possible class equation for G . [Use Exercise 36, Section 3.1.]
- For $n = 3, 4, 6$ and 7 make lists of the partitions of n and give representatives for the corresponding conjugacy classes of S_n .
- Prove that $Z(S_n) = 1$ for all $n \geq 3$.
- Show that $|C_{S_n}((1\ 2)(3\ 4))| = 8 \cdot (n-4)!$ for all $n \geq 4$. Determine the elements in this centralizer explicitly.
- Let σ be the 5-cycle $(1\ 2\ 3\ 4\ 5)$ in S_5 . In each of (a) to (c) find an explicit element $\tau \in S_5$ which accomplishes the specified conjugation:
(a) $\tau\sigma\tau^{-1} = \sigma^2$
(b) $\tau\sigma\tau^{-1} = \sigma^{-1}$
(c) $\tau\sigma\tau^{-1} = \sigma^{-2}$.
- In each of (a) – (d) determine whether σ_1 and σ_2 are conjugate. If they are, give an explicit permutation τ such that $\tau\sigma_1\tau^{-1} = \sigma_2$.
(a) $\sigma_1 = (1\ 2)(3\ 4\ 5)$ and $\sigma_2 = (1\ 2\ 3)(4\ 5)$
(b) $\sigma_1 = (1\ 5)(3\ 7\ 2)(10\ 6\ 8\ 11)$ and $\sigma_2 = (3\ 7\ 5\ 10)(4\ 9)(13\ 11\ 2)$
(c) $\sigma_1 = (1\ 5)(3\ 7\ 2)(10\ 6\ 8\ 11)$ and $\sigma_2 = \sigma_1^3$
(d) $\sigma_1 = (1\ 3)(2\ 4\ 6)$ and $\sigma_2 = (3\ 5)(2\ 4)(5\ 6)$.
- Find a representative for each conjugacy class of elements of order 4 in S_8 and in S_{12} .
- Find all finite groups which have exactly two conjugacy classes.
- In Exercise 1 of Section 2 two labellings of the elements $\{1, a, b, c\}$ of the Klein 4-group V were chosen to give two versions of the left regular representation of V into S_4 . Let π_1 be the version of regular representation obtained in part (a) of that exercise and let π_2 be the version obtained via the labelling in part (b). Let $\tau = (2\ 4)$. Show that $\tau \circ \pi_1(g) \circ \tau^{-1} = \pi_2(g)$ for each $g \in V$ (i.e., conjugation by τ sends the image of π_1 to the image of π_2 elementwise).

15. Find an element of S_8 which conjugates the subgroup of S_8 obtained in part (a) of Exercise 3, Section 2 to the subgroup of S_8 obtained in part (b) of that same exercise (both of these subgroups are isomorphic to D_8).
16. Find an element of S_4 which conjugates the subgroup of S_4 obtained in part (a) of Exercise 5, Section 2 to the subgroup of S_4 obtained in part (b) of that same exercise (both of these subgroups are isomorphic to D_8).
17. Let A be a nonempty set and let X be any subset of S_A . Let

$$F(X) = \{a \in A \mid \sigma(a) = a \text{ for all } \sigma \in X\} \quad \text{— the fixed set of } X.$$

- Let $M(X) = A - F(X)$ be the elements which are *moved* by some element of X . Let $D = \{\sigma \in S_A \mid |M(\sigma)| < \infty\}$. Prove that D is a normal subgroup of S_A .
18. Let A be a set, let H be a subgroup of S_A and let $F(H)$ be the fixed points of H on A as defined in the preceding exercise. Prove that if $\tau \in N_{S_A}(H)$ then τ stabilizes the set $F(H)$ and its complement $A - F(H)$.
 19. Assume H is a normal subgroup of G , $\mathcal{K}$ is a conjugacy class of G contained in H and $x \in \mathcal{K}$. Prove that $\mathcal{K}$ is a union of k conjugacy classes of equal size in H , where $k = |G : H C_G(x)|$. Deduce that a conjugacy class in S_n which consists of even permutations is either a single conjugacy class under the action of A_n or is a union of two classes of the same size in A_n . [Let $A = C_G(x)$ and $B = H$ so $A \cap B = C_H(x)$. Draw the lattice diagram associated to the Second Isomorphism Theorem and interpret the appropriate indices. See also Exercise 9, Section 1.]
 20. Let $\sigma \in A_n$. Show that all elements in the conjugacy class of σ in S_n (i.e., all elements of the same cycle type as σ) are conjugate in A_n if and only if σ commutes with an odd permutation. [Use the preceding exercise.]
 21. Let $\mathcal{K}$ be a conjugacy class in S_n and assume that $\mathcal{K} \subseteq A_n$. Show $\sigma \in S_n$ does *not* commute with any odd permutation if and only if the cycle type of σ consists of distinct odd integers. Deduce that $\mathcal{K}$ consists of two conjugacy classes in A_n if and only if the cycle type of an element of $\mathcal{K}$ consists of distinct odd integers. [Assume first that $\sigma \in \mathcal{K}$ does not commute with any odd permutation. Observe that σ commutes with each individual cycle in its cycle decomposition — use this to show that all its cycles must be of odd length. If two cycles have the same odd length, k , find a product of k transpositions which interchanges them and commutes with σ . Conversely, if the cycle type of σ consists of distinct integers, prove that σ commutes *only* with the group generated by the cycles in its cycle decomposition.]
 22. Show that if n is odd then the set of all n -cycles consists of two conjugacy classes of equal size in A_n .
 23. Recall (cf. Exercise 16, Section 2.4) that a proper subgroup M of G is called *maximal* if whenever $M \leq H \leq G$, either $H = M$ or $H = G$. Prove that if M is a maximal subgroup of G then either $N_G(M) = M$ or $N_G(M) = G$. Deduce that if M is a maximal subgroup of G that is not normal in G then the number of nonidentity elements of G that are contained in conjugates of M is at most $(|M| - 1)|G : M|$.
 24. Assume H is a proper subgroup of the finite group G . Prove $G \neq \bigcup_{g \in G} Hg^{-1}$, i.e., G is not the union of the conjugates of any proper subgroup. [Put H in some maximal subgroup and use the preceding exercise.]
 25. Let $G = GL_2(\mathbb{C})$ and let $H = \left\{ \begin{pmatrix} a & b \\ 0 & c \end{pmatrix} \mid a, b, c \in \mathbb{C}, ac \neq 0 \right\}$. Prove that every element of G is conjugate to some element of the subgroup H and deduce that G is the union of

conjugates of H . [Show that every element of $GL_2(\mathbb{C})$ has an eigenvector.]

26. Let G be a transitive permutation group on the finite set A with $|A| > 1$. Show that there is some $\sigma \in G$ such that $\sigma(a) \neq a$ for all $a \in A$ (such an element σ is called *fixed point free*).
27. Let $g_1, g_2, \dots, g_r$ be representatives of the conjugacy classes of the finite group G and assume these elements pairwise commute. Prove that G is abelian.
28. Let p and q be primes with $p < q$. Prove that a non-abelian group G of order pq has a nonnormal subgroup of index q , so that there exists an injective homomorphism into S_q . Deduce that G is isomorphic to a subgroup of the normalizer in S_q of the cyclic group generated by the q -cycle $(1\ 2\ \dots\ q)$.
29. Let p be a prime and let G be a group of order p^α . Prove that G has a subgroup of order p^β , for every β with $0 \leq \beta \leq \alpha$. [Use Theorem 8 and induction on α .]
30. If G is a group of odd order, prove for any nonidentity element $x \in G$ that x and x^{-1} are not conjugate in G .
31. Using the usual generators and relations for the dihedral group D_{2n} (cf. Section 1.2) show that for $n = 2k$ an even integer the conjugacy classes in D_{2n} are the following: $\{1\}$, $\{r^k\}$, $\{r^{\pm 1}\}$, $\{r^{\pm 2}\}$, $\dots$, $\{r^{\pm(k-1)}\}$, $\{sr^{2b} \mid b = 1, \dots, k\}$ and $\{sr^{2b-1} \mid b = 1, \dots, k\}$. Give the class equation for D_{2n} .
32. For $n = 2k + 1$ an odd integer show that the conjugacy classes in D_{2n} are $\{1\}$, $\{r^{\pm 1}\}$, $\{r^{\pm 2}\}$, $\dots$, $\{r^{\pm k}\}$, $\{sr^b \mid b = 1, \dots, n\}$. Give the class equation for D_{2n} .
33. This exercise gives a formula for the size of each conjugacy class in S_n . Let σ be a permutation in S_n and let $m_1, m_2, \dots, m_s$ be the *distinct* integers which appear in the cycle type of σ (including 1-cycles). For each $i \in \{1, 2, \dots, s\}$ assume σ has k_i cycles of length m_i (so that $\sum_{i=1}^s k_i m_i = n$). Prove that the number of conjugates of σ is

$$\frac{n!}{(k_1! m_1^{k_1})(k_2! m_2^{k_2}) \dots (k_s! m_s^{k_s})}.$$

[See Exercises 6 and 7 in Section 1.3 where this formula was given in some special cases.]

34. Prove that if p is a prime and P is a subgroup of S_p of order p , then $|N_{S_p}(P)| = p(p-1)$. [Argue that every conjugate of P contains exactly $p-1$ p -cycles and use the formula for the number of p -cycles to compute the index of $N_{S_p}(P)$ in S_p .]
35. Let p be a prime. Find a formula for the number of conjugacy classes of elements of order p in S_n (using the greatest integer function).
36. Let $\pi : G \rightarrow S_G$ be the left regular representation afforded by the action of G on itself by left multiplication. For each $g \in G$ denote the permutation $\pi(g)$ by σ_g , so that $\sigma_g(x) = gx$ for all $x \in G$. Let $\lambda : G \rightarrow S_G$ be the permutation representation afforded by the corresponding right action of G on itself, and for each $h \in G$ denote the permutation $\lambda(h)$ by τ_h . Thus $\tau_h(x) = xh^{-1}$ for all $x \in G$ (λ is called the *right regular representation* of G).
 - (a) Prove that σ_g and τ_h commute for all $g, h \in G$. (Thus the centralizer in S_G of $\pi(G)$ contains the subgroup $\lambda(G)$, which is isomorphic to G).
 - (b) Prove that $\sigma_g = \tau_g$ if and only if g is an element of order 1 or 2 in the center of G .
 - (c) Prove that $\sigma_g = \tau_h$ if and only if g and h lie in the center of G . Deduce that $\pi(G) \cap \lambda(G) = \pi(Z(G)) = \lambda(Z(G))$.

4.4 AUTOMORPHISMS

Definition. Let G be a group. An isomorphism from G onto itself is called an *automorphism* of G . The set of all automorphisms of G is denoted by $\text{Aut}(G)$.

We leave as an exercise the simple verification that $\text{Aut}(G)$ is a group under composition of automorphisms, the *automorphism group* of G (composition of automorphisms is defined since the domain and range of each automorphism is the same). Notice that automorphisms of a group G are, in particular, permutations of the set G so $\text{Aut}(G)$ is a subgroup of S_G .

One of the most important examples of an automorphism of a group G is provided by conjugation by a fixed element in G . The next result discusses this in a slightly more general context.

Proposition 13. Let H be a normal subgroup of the group G . Then G acts by conjugation on H as automorphisms of H . More specifically, the action of G on H by conjugation is defined for each $g \in G$ by

$$h \mapsto ghg^{-1} \quad \text{for each } h \in H.$$

For each $g \in G$, conjugation by g is an automorphism of H . The permutation representation afforded by this action is a homomorphism of G into $\text{Aut}(H)$ with kernel $C_G(H)$. In particular, $G/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$.

Proof: (cf. Exercise 17, Section 1.7) Let φ_g be conjugation by g . Note that because g normalizes H , φ_g maps H to itself. Since we have already seen that conjugation defines an action, it follows that $\varphi_1 = 1$ (the identity map on H) and $\varphi_a \circ \varphi_b = \varphi_{ab}$ for all $a, b \in G$. Thus each φ_g gives a bijection from H to itself since it has a 2-sided inverse $\varphi_{g^{-1}}$. Each φ_g is a homomorphism from H to H because

$$\varphi_g(hk) = g(hk)g^{-1} = gh(gg^{-1})kg^{-1} = (ghg^{-1})(gkg^{-1}) = \varphi_g(h)\varphi_g(k)$$

for all $h, k \in H$. This proves that conjugation by any fixed element of G defines an automorphism of H .

By the preceding remark, the permutation representation $\psi : G \rightarrow S_H$ defined by $\psi(g) = \varphi_g$ (which we have already proved is a homomorphism) has image contained in the subgroup $\text{Aut}(H)$ of S_H . Finally,

$$\begin{aligned} \ker \psi &= \{g \in G \mid \varphi_g = \text{id}\} \\ &= \{g \in G \mid ghg^{-1} = h \text{ for all } h \in H\} \\ &= C_G(H). \end{aligned}$$

The First Isomorphism Theorem implies the final statement of the proposition.

Proposition 13 shows that a group acts by conjugation on a normal subgroup as *structure preserving* permutations, i.e., as automorphisms. In particular, this action must send subgroups to subgroups, elements of order n to elements of order n , etc. Two specific applications of this proposition are described in the next two corollaries.

Corollary 14. If K is any subgroup of the group G and $g \in G$, then $K \cong gKg^{-1}$. Conjugate elements and conjugate subgroups have the same order.

Proof: Letting $G = H$ in the proposition shows that conjugation by $g \in G$ is an automorphism of G , from which the corollary follows.

Corollary 15. For any subgroup H of a group G , the quotient group $N_G(H)/C_G(H)$ is isomorphic to a subgroup of $\text{Aut}(H)$. In particular, $G/Z(G)$ is isomorphic to a subgroup of $\text{Aut}(G)$.

Proof: Since H is a normal subgroup of the group $N_G(H)$, Proposition 13 (applied with $N_G(H)$ playing the role of G) implies the first assertion. The second assertion is the special case when $H = G$, in which case $N_G(G) = G$ and $C_G(G) = Z(G)$.

Definition. Let G be a group and let $g \in G$. Conjugation by g is called an *inner automorphism* of G and the subgroup of $\text{Aut}(G)$ consisting of all inner automorphisms is denoted by $\text{Inn}(G)$.

Note that the collection of inner automorphisms of G is in fact a subgroup of $\text{Aut}(G)$ and that by Corollary 15, $\text{Inn}(G) \cong G/Z(G)$. Note also that if H is a normal subgroup of G , conjugation by an element of G when restricted to H is an automorphism of H but need not be an inner automorphism of H (as we shall see).

Examples

- (1) A group G is abelian if and only if every inner automorphism is trivial. If H is an abelian normal subgroup of G and H is not contained in $Z(G)$, then there is some $g \in G$ such that conjugation by g restricted to H is not an inner automorphism of H . An explicit example of this is $G = A_4$, H is the Klein 4-group in G and g is any 3-cycle.
- (2) Since $Z(Q_8) = \langle -1 \rangle$ we have $\text{Inn}(Q_8) \cong V_4$.
- (3) Since $Z(D_8) = \langle r^2 \rangle$ we have $\text{Inn}(D_8) \cong V_4$.
- (4) Since for all $n \geq 3$, $Z(S_n) = 1$ we have $\text{Inn}(S_n) \cong S_n$.

Corollary 15 shows that any information we have about the automorphism group of a subgroup H of a group G translates into information about $N_G(H)/C_G(H)$. For example, if $H \cong Z_2$, then since H has unique elements of orders 1 and 2, Corollary 14 forces $\text{Aut}(H) = 1$. Thus if $H \cong Z_2$, $N_G(H) = C_G(H)$; if in addition H is a normal subgroup of G , then $H \leq Z(G)$ (cf. Exercise 10, Section 2.2).

Although the preceding example was fairly trivial, it illustrates that the action of G by conjugation on a *normal* subgroup H can be restricted by knowledge of the automorphism group of H . This in turn can be used to investigate the structure of G and will lead to some classification theorems when we consider semidirect products in Section 5.5.

A notion which will be used in later sections most naturally warrants introduction here:

Definition. A subgroup H of a group G is called *characteristic* in G , denoted $H \text{ char } G$, if every automorphism of G maps H to itself, i.e., $\sigma(H) = H$ for all $\sigma \in \text{Aut}(G)$.

Results concerning characteristic subgroups which we shall use later (and whose proofs are relegated to the exercises) are

- (1) characteristic subgroups are normal,
- (2) if H is the unique subgroup of G of a given order, then H is characteristic in G , and
- (3) if $K \text{ char } H$ and $H \trianglelefteq G$, then $K \trianglelefteq G$ (so although “normality” is not a transitive property (i.e., a normal subgroup of a normal subgroup need not be normal), a characteristic subgroup of a normal subgroup is normal).

Thus we may think of characteristic subgroups as “strongly normal” subgroups. For example, property (2) and Theorem 2.7 imply that every subgroup of a cyclic group is characteristic.

We close this section with some results on automorphism groups of specific groups.

Proposition 16. The automorphism group of the cyclic group of order n is isomorphic to $(\mathbb{Z}/n\mathbb{Z})^\times$, an abelian group of order $\varphi(n)$ (where φ is Euler’s function).

Proof: Let x be a generator of the cyclic group Z_n . If $\psi \in \text{Aut}(Z_n)$, then $\psi(x) = x^a$ for some $a \in \mathbb{Z}$ and the integer a uniquely determines ψ . Denote this automorphism by ψ_a . As usual, since $|x| = n$, the integer a is only defined mod n . Since ψ_a is an automorphism, x and x^a must have the same order, hence $(a, n) = 1$. Furthermore, for every a relatively prime to n , the map $x \mapsto x^a$ is an automorphism of Z_n . Hence we have a surjective map

$$\begin{aligned}\Psi : \text{Aut}(Z_n) &\rightarrow (\mathbb{Z}/n\mathbb{Z})^\times \\ \psi_a &\mapsto a \pmod{n}.\end{aligned}$$

The map Ψ is a homomorphism because

$$\psi_a \circ \psi_b(x) = \psi_a(x^b) = (x^b)^a = x^{ab} = \psi_{ab}(x)$$

for all $\psi_a, \psi_b \in \text{Aut}(Z_n)$, so that

$$\Psi(\psi_a \circ \psi_b) = \Psi(\psi_{ab}) = ab \pmod{n} = \Psi(\psi_a)\Psi(\psi_b).$$

Finally, Ψ is clearly injective, hence is an isomorphism.

A complete description of the isomorphism type of $\text{Aut}(Z_n)$ is given at the end of Section 9.5.

Example

Assume G is a group of order pq , where p and q are primes (not necessarily distinct) with $p \leq q$. If $p \nmid q-1$, we prove G is abelian.

If $Z(G) \neq 1$, Lagrange’s Theorem forces $G/Z(G)$ to be cyclic, hence G is abelian by Exercise 36, Section 3.1. Hence we may assume $Z(G) = 1$.

If every nonidentity element of G has order p , then the centralizer of every nonidentity element has index q , so the class equation for G reads

$$pq = 1 + kq.$$

This is impossible since q divides pq and kq but not 1. Thus G contains an element, x , of order q .

Let $H = \langle x \rangle$. Since H has index p and p is the smallest prime dividing $|G|$, the subgroup H is normal in G by Corollary 5. Since $Z(G) = 1$, we must have $C_G(H) = H$. Thus $G/H = N_G(H)/C_G(H)$ is a group of order p isomorphic to a subgroup of $\text{Aut}(H)$ by Corollary 15. But by Proposition 16, $\text{Aut}(H)$ has order $\varphi(q) = q - 1$, which by Lagrange's Theorem would imply $p \mid q - 1$, contrary to assumption. This shows that G must be abelian.

One can check that every group of order pq , where p and q are distinct primes with $p < q$ and $p \nmid q - 1$ is *cyclic* (see the exercises). This is the first instance where there is a unique isomorphism type of group whose order is *composite*. For instance, every group of order 15 is cyclic.

The next proposition summarizes some results on automorphism groups of known groups and will be proved later. Part 3 of this proposition illustrates how the theory of vector spaces comes into play in group theory.

Proposition 17.

- (1) If p is an odd prime and $n \in \mathbb{Z}^+$, then the automorphism group of the cyclic group of order p is cyclic of order $p - 1$. More generally, the automorphism group of the cyclic group of order p^n is cyclic of order $p^{n-1}(p-1)$ (cf. Corollary 20, Section 9.5).
- (2) For all $n \geq 3$ the automorphism group of the cyclic group of order 2^n is isomorphic to $Z_2 \times Z_{2^{n-2}}$, and in particular is not cyclic but has a cyclic subgroup of index 2 (cf. Corollary 20, Section 9.5).
- (3) Let p be a prime and let V be an abelian group (written additively) with the property that $pv = 0$ for all $v \in V$. If $|V| = p^n$, then V is an n -dimensional vector space over the field $\mathbb{F}_p = \mathbb{Z}/p\mathbb{Z}$. The automorphisms of V are precisely the nonsingular linear transformations from V to itself, that is

$$\text{Aut}(V) \cong GL(V) \cong GL_n(\mathbb{F}_p).$$

In particular, the order of $\text{Aut}(V)$ is as given in Section 1.4 (cf. the examples in Sections 10.2 and 11.1).

- (4) For all $n \neq 6$ we have $\text{Aut}(S_n) = \text{Inn}(S_n) \cong S_n$ (cf. Exercise 18). For $n = 6$ we have $|\text{Aut}(S_6) : \text{Inn}(S_6)| = 2$ (cf. the following Exercise 19 and also Exercise 10 in Section 6.3).
- (5) $\text{Aut}(D_8) \cong D_8$ and $\text{Aut}(Q_8) \cong S_4$ (cf. the following Exercises 4 and 5 and also Exercise 9 in Section 6.3).

The group V described in Part 3 of the proposition is called the *elementary abelian* group of order p^n (we shall see in Chapter 5 that it is uniquely determined up to isomorphism by p and n). The Klein 4-group, V_4 , is the elementary abelian group of order 4. This proposition asserts that

$$\text{Aut}(V_4) \cong GL_2(\mathbb{F}_2).$$

By the exercises in Section 1.4, the latter group has order 6. But $\text{Aut}(V_4)$ permutes the 3 nonidentity elements of V_4 , and this action of $\text{Aut}(V_4)$ on $V_4 - \{1\}$ gives an injective permutation representation of $\text{Aut}(V_4)$ into S_3 . By order considerations, the homomorphism is onto, so

$$\text{Aut}(V_4) \cong GL_2(\mathbb{F}_2) \cong S_3.$$

Note that V_4 is abelian, so $\text{Inn}(V_4) = 1$.

For any prime p , the elementary abelian group of order p^2 is $Z_p \times Z_p$. Its automorphism group, $GL_2(\mathbb{F}_p)$, has order $p(p-1)^2(p+1)$. Thus Corollary 9 implies that for p a prime

$$\text{if } |P| = p^2, \quad |\text{Aut}(P)| = p(p-1) \text{ or } p(p-1)^2(p+1)$$

according to whether P is cyclic or elementary abelian, respectively.

Example

Suppose G is a group of order $45 = 3^2 \cdot 5$ with a normal subgroup P of order 3^2 . We show that G is necessarily abelian.

The quotient $G/C_G(P)$ is isomorphic to a subgroup of $\text{Aut}(P)$ by Corollary 15, and $\text{Aut}(P)$ has order 6 or 48 (according to whether P is cyclic or elementary abelian, respectively) by the preceding paragraph. On the other hand, since the order of P is the square of a prime, P is an abelian group, hence $P \leq C_G(P)$. It follows that $|C_G(P)|$ is divisible by 9, which implies $|G/C_G(P)|$ is 1 or 5. Together these imply $|G/C_G(P)| = 1$, i.e., $C_G(P) = G$ and $P \leq Z(G)$. Since then $G/Z(G)$ is cyclic, G must be an abelian group.

EXERCISES

Let G be a group.

1. If $\sigma \in \text{Aut}(G)$ and φ_g is conjugation by g prove $\sigma \varphi_g \sigma^{-1} = \varphi_{\sigma(g)}$. Deduce that $\text{Inn}(G) \leq \text{Aut}(G)$. (The group $\text{Aut}(G)/\text{Inn}(G)$ is called the *outer automorphism group* of G .)
2. Prove that if G is an abelian group of order pq , where p and q are distinct primes, then G is cyclic. [Use Cauchy's Theorem to produce elements of order p and q and consider the order of their product.]
3. Prove that under any automorphism of D_8 , r has at most 2 possible images and s has at most 4 possible images (r and s are the usual generators — cf. Section 1.2). Deduce that $|\text{Aut}(D_8)| \leq 8$.
4. Use arguments similar to those in the preceding exercise to show $|\text{Aut}(Q_8)| \leq 24$.
5. Use the fact that $D_8 \trianglelefteq D_{16}$ to prove that $\text{Aut}(D_8) \cong D_8$.
6. Prove that characteristic subgroups are normal. Give an example of a normal subgroup that is not characteristic.
7. If H is the unique subgroup of a given order in a group G prove H is characteristic in G .
8. Let G be a group with subgroups H and K with $H \leq K$.
 - (a) Prove that if H is characteristic in K and K is normal in G then H is normal in G .
 - (b) Prove that if H is characteristic in K and K is characteristic in G then H is characteristic in G . Use this to prove that the Klein 4-group V_4 is characteristic in S_4 .
 - (c) Give an example to show that if H is normal in K and K is characteristic in G then H need not be normal in G .

9. If r, s are the usual generators for the dihedral group D_{2n} , use the preceding two exercises to deduce that every subgroup of $\langle r \rangle$ is normal in D_{2n} .
10. Let G be a group, let A be an abelian normal subgroup of G , and write $\bar{G} = G/A$. Show that $\bar{G}$ acts (on the left) by conjugation on A by $\bar{g} \cdot a = gag^{-1}$, where g is any representative of the coset $\bar{g}$ (in particular, show that this action is well defined). Give an explicit example to show that this action is not well defined if A is non-abelian.
11. If p is a prime and P is a subgroup of S_p of order p , prove $N_{S_p}(P)/C_{S_p}(P) \cong \text{Aut}(P)$. [Use Exercise 34, Section 3.]
12. Let G be a group of order 3825. Prove that if H is a normal subgroup of order 17 in G then $H \leq Z(G)$.
13. Let G be a group of order 203. Prove that if H is a normal subgroup of order 7 in G then $H \leq Z(G)$. Deduce that G is abelian in this case.
14. Let G be a group of order 1575. Prove that if H is a normal subgroup of order 9 in G then $H \leq Z(G)$.
15. Prove that each of the following (multiplicative) groups is cyclic: $(\mathbb{Z}/5\mathbb{Z})^\times$, $(\mathbb{Z}/9\mathbb{Z})^\times$ and $(\mathbb{Z}/18\mathbb{Z})^\times$.
16. Prove that $(\mathbb{Z}/24\mathbb{Z})^\times$ is an elementary abelian group of order 8. (We shall see later that $(\mathbb{Z}/n\mathbb{Z})^\times$ is an elementary abelian group if and only if $n \mid 24$.)
17. Let $G = \langle x \rangle$ be a cyclic group of order n . For $n = 2, 3, 4, 5, 6$ write out the elements of $\text{Aut}(G)$ explicitly (by Proposition 16 above we know $\text{Aut}(G) \cong (\mathbb{Z}/n\mathbb{Z})^\times$, so for each element $a \in (\mathbb{Z}/n\mathbb{Z})^\times$, write out explicitly what the automorphism ψ_a does to the elements $\{1, x, x^2, \dots, x^{n-1}\}$ of G).
18. This exercise shows that for $n \neq 6$ every automorphism of S_n is inner. Fix an integer $n \geq 2$ with $n \neq 6$.
 - (a) Prove that the automorphism group of a group G permutes the conjugacy classes of G , i.e., for each $\sigma \in \text{Aut}(G)$ and each conjugacy class $\mathcal{K}$ of G the set $\sigma(\mathcal{K})$ is also a conjugacy class of G .
 - (b) Let $\mathcal{K}$ be the conjugacy class of transpositions in S_n and let $\mathcal{K}'$ be the conjugacy class of any element of order 2 in S_n that is not a transposition. Prove that $|\mathcal{K}| \neq |\mathcal{K}'|$. Deduce that any automorphism of S_n sends transpositions to transpositions. [See Exercise 33 in Section 3.]
 - (c) Prove that for each $\sigma \in \text{Aut}(S_n)$

$$\sigma : (1\ 2) \mapsto (a\ b_2), \quad \sigma : (1\ 3) \mapsto (a\ b_3), \quad \dots, \quad \sigma : (1\ n) \mapsto (a\ b_n)$$
 for some distinct integers $a, b_2, b_3, \dots, b_n \in \{1, 2, \dots, n\}$.
 - (d) Show that $(1\ 2), (1\ 3), \dots, (1\ n)$ generate S_n and deduce that any automorphism of S_n is uniquely determined by its action on these elements. Use (c) to show that S_n has at most $n!$ automorphisms and conclude that $\text{Aut}(S_n) = \text{Inn}(S_n)$ for $n \neq 6$.
19. This exercise shows that $|\text{Aut}(S_6) : \text{Inn}(S_6)| \leq 2$ (Exercise 10 in Section 6.3 shows that equality holds by exhibiting an automorphism of S_6 that is not inner).
 - (a) Let $\mathcal{K}$ be the conjugacy class of transpositions in S_6 and let $\mathcal{K}'$ be the conjugacy class of any element of order 2 in S_6 that is not a transposition. Prove that $|\mathcal{K}| \neq |\mathcal{K}'|$ unless $\mathcal{K}'$ is the conjugacy class of products of three disjoint transpositions. Deduce that $\text{Aut}(S_6)$ has a subgroup of index at most 2 which sends transpositions to transpositions.
 - (b) Prove that $|\text{Aut}(S_6) : \text{Inn}(S_6)| \leq 2$. [Follow the same steps as in (c) and (d) of the preceding exercise to show that any automorphism that sends transpositions to transpositions is inner.]

The next exercise introduces a subgroup, $J(P)$, which (like the center of P) is defined for an arbitrary finite group P (although in most applications P is a group whose order is a power of a prime). This subgroup was defined by J. Thompson in 1964 and it now plays a pivotal role in the study of finite groups, in particular, in the classification of finite simple groups.

20. For any finite group P let $d(P)$ be the minimum number of generators of P (so, for example, $d(P) = 1$ if and only if P is a nontrivial cyclic group and $d(Q_8) = 2$). Let $m(P)$ be the maximum of the integers $d(A)$ as A runs over all *abelian* subgroups of P (so, for example, $m(Q_8) = 1$ and $m(D_8) = 2$). Define

$$J(P) = \langle A \mid A \text{ is an abelian subgroup of } P \text{ with } d(A) = m(P) \rangle.$$

($J(P)$ is called the *Thompson subgroup* of P .)

- Prove that $J(P)$ is a characteristic subgroup of P .
- For each of the following groups P list all abelian subgroups A of P that satisfy $d(A) = m(P)$: Q_8 , D_8 , D_{16} and QD_{16} (where QD_{16} is the quasidihedral group of order 16 defined in Exercise 11 of Section 2.5). [Use the lattices of subgroups for these groups in Section 2.5.]
- Show that $J(Q_8) = Q_8$, $J(D_8) = D_8$, $J(D_{16}) = D_{16}$ and $J(QD_{16})$ is a dihedral subgroup of order 8 in QD_{16} .
- Prove that if $Q \leq P$ and $J(P)$ is a subgroup of Q , then $J(P) = J(Q)$. Deduce that if P is a subgroup (not necessarily normal) of the finite group G and $J(P)$ is contained in some subgroup Q of P such that $Q \leq G$, then $J(P) \leq G$.

4.5 SYLOW'S THEOREM

In this section we prove a partial converse to Lagrange's Theorem and derive numerous consequences, some of which will lead to classification theorems in the next chapter.

Definition. Let G be a group and let p be a prime.

- A group of order p^α for some $\alpha \geq 1$ is called a *p-group*. Subgroups of G which are *p-groups* are called *p-subgroups*.
- If G is a group of order $p^\alpha m$, where $p \nmid m$, then a subgroup of order p^α is called a *Sylow p-subgroup* of G .
- The set of Sylow *p*-subgroups of G will be denoted by $Syl_p(G)$ and the number of Sylow *p*-subgroups of G will be denoted by $n_p(G)$ (or just n_p when G is clear from the context).

Theorem 18. (Sylow's Theorem) Let G be a group of order $p^\alpha m$, where p is a prime not dividing m .

- Sylow *p*-subgroups of G exist, i.e., $Syl_p(G) \neq \emptyset$.
- If P is a Sylow *p*-subgroup of G and Q is any *p*-subgroup of G , then there exists $g \in G$ such that $Q \leq gPg^{-1}$, i.e., Q is contained in some conjugate of P . In particular, any two Sylow *p*-subgroups of G are conjugate in G .
- The number of Sylow *p*-subgroups of G is of the form $1 + kp$, i.e.,

$$n_p \equiv 1 \pmod{p}.$$

Further, n_p is the index in G of the normalizer $N_G(P)$ for any Sylow *p*-subgroup P , hence n_p divides m .

We first prove the following lemma:

Lemma 19. Let $P \in \text{Syl}_p(G)$. If Q is any p -subgroup of G , then $Q \cap N_G(P) = Q \cap P$.

Proof: Let $H = N_G(P) \cap Q$. Since $P \leq N_G(P)$ it is clear that $P \cap Q \leq H$, so we must prove the reverse inclusion. Since by definition $H \leq Q$, this is equivalent to showing $H \leq P$. We do this by demonstrating that PH is a p -subgroup of G containing both P and H ; but P is a p -subgroup of G of largest possible order, so we must have $PH = P$, i.e., $H \leq P$.

Since $H \leq N_G(P)$, by Corollary 15 in Section 3.2, PH is a subgroup. By Proposition 13 in the same section

$$|PH| = \frac{|P||H|}{|P \cap H|}.$$

All the numbers in the above quotient are powers of p , so PH is a p -group. Moreover, P is a subgroup of PH so the order of PH is divisible by p^α , the largest power of p which divides $|G|$. These two facts force $|PH| = p^\alpha = |P|$. This in turn implies $P = PH$ and $H \leq P$. This establishes the lemma.

Proof of Sylow's Theorem (1) Proceed by induction on $|G|$. If $|G| = 1$, there is nothing to prove. Assume inductively the existence of Sylow p -subgroups for all groups of order less than $|G|$.

If p divides $|Z(G)|$, then by Cauchy's Theorem for abelian groups (Proposition 21, Section 3.4) $Z(G)$ has a subgroup, N , of order p . Let $\bar{G} = G/N$, so that $|\bar{G}| = p^{\alpha-1}m$. By induction, $\bar{G}$ has a subgroup $\bar{P}$ of order $p^{\alpha-1}$. If we let P be the subgroup of G containing N such that $P/N = \bar{P}$ then $|P| = |P/N| \cdot |N| = p^\alpha$ and P is a Sylow p -subgroup of G . We are reduced to the case when p does not divide $|Z(G)|$.

Let $g_1, g_2, \dots, g_r$ be representatives of the distinct non-central conjugacy classes of G . The class equation for G is

$$|G| = |Z(G)| + \sum_{i=1}^r |G : C_G(g_i)|.$$

If $p \mid |G : C_G(g_i)|$ for all i , then since $p \mid |G|$, we would also have $p \mid |Z(G)|$, a contradiction. Thus for some i , p does not divide $|G : C_G(g_i)|$. For this i let $H = C_G(g_i)$ so that

$$|H| = p^\alpha k, \quad \text{where } p \nmid k.$$

Since $g_i \notin Z(G)$, $|H| < |G|$. By induction, H has a Sylow p -subgroup, P , which of course is also a subgroup of G . Since $|P| = p^\alpha$, P is a Sylow p -subgroup of G . This completes the induction and establishes (1).

Before proving (2) and (3) we make some calculations. By (1) there exists a Sylow p -subgroup, P , of G . Let

$$\{P_1, P_2, \dots, P_r\} = \mathcal{S}$$

be the set of all conjugates of P (i.e., $\mathcal{S} = \{gPg^{-1} \mid g \in G\}$) and let Q be any p -subgroup of G . By definition of $\mathcal{S}$, G , hence also Q , acts by conjugation on $\mathcal{S}$. Write $\mathcal{S}$ as a disjoint union of orbits under this action by Q :

$$\mathcal{S} = \mathcal{O}_1 \cup \mathcal{O}_2 \cup \dots \cup \mathcal{O}_s$$

where $r = |\mathcal{O}_1| + \dots + |\mathcal{O}_s|$. Keep in mind that r does not depend on Q but the number of Q -orbits s does (note that by definition, G has only one orbit on S but a subgroup Q of G may have more than one orbit). Renumber the elements of S if necessary so that the first s elements of S are representatives of the Q -orbits: $P_i \in \mathcal{O}_i$, $1 \leq i \leq s$. It follows from Proposition 2 that $|\mathcal{O}_i| = |Q : N_Q(P_i)|$. By definition, $N_Q(P_i) = N_G(P_i) \cap Q$ and by Lemma 19, $N_G(P_i) \cap Q = P_i \cap Q$. Combining these two facts gives

$$|\mathcal{O}_i| = |Q : P_i \cap Q|, \quad 1 \leq i \leq s. \quad (4.1)$$

We are now in a position to prove that $r \equiv 1 \pmod{p}$. Since Q was arbitrary we may take $Q = P_1$ above, so that (1) gives

$$|\mathcal{O}_1| = 1.$$

Now, for all $i > 1$, $P_1 \neq P_i$, so $P_1 \cap P_i < P_1$. By (1)

$$|\mathcal{O}_i| = |P_1 : P_1 \cap P_i| > 1, \quad 2 \leq i \leq s.$$

Since P_1 is a p -group, $|P_1 : P_1 \cap P_i|$ must be a power of p , so that

$$p \mid |\mathcal{O}_i|, \quad 2 \leq i \leq s.$$

Thus

$$r = |\mathcal{O}_1| + (|\mathcal{O}_2| + \dots + |\mathcal{O}_s|) \equiv 1 \pmod{p}.$$

We now prove parts (2) and (3). Let Q be any p -subgroup of G . Suppose Q is not contained in P_i for any $i \in \{1, 2, \dots, r\}$ (i.e., $Q \not\leq gPg^{-1}$ for any $g \in G$). In this situation, $Q \cap P_i < Q$ for all i , so by (1)

$$|\mathcal{O}_i| = |Q : Q \cap P_i| > 1, \quad 1 \leq i \leq s.$$

Thus $p \mid |\mathcal{O}_i|$ for all i , so p divides $|\mathcal{O}_1| + \dots + |\mathcal{O}_s| = r$. This contradicts the fact that $r \equiv 1 \pmod{p}$ (remember, r does not depend on the choice of Q). This contradiction proves $Q \leq gPg^{-1}$ for some $g \in G$.

To see that all Sylow p -subgroups of G are conjugate, let Q be any Sylow p -subgroup of G . By the preceding argument, $Q \leq gPg^{-1}$ for some $g \in G$. Since $|gPg^{-1}| = |Q| = p^\alpha$, we must have $gPg^{-1} = Q$. This establishes part (2) of the theorem. In particular, $S = \text{Syl}_p(G)$ since every Sylow p -subgroup of G is conjugate to P , and so $n_p = r \equiv 1 \pmod{p}$, which is the first part of (3).

Finally, since all Sylow p -subgroups are conjugate, Proposition 6 shows that

$$n_p = |G : N_G(P)| \quad \text{for any } P \in \text{Syl}_p(G),$$

completing the proof of Sylow's Theorem.

Note that the conjugacy part of Sylow's Theorem together with Corollary 14 shows that any two Sylow p -subgroups of a group (for the same prime p) are isomorphic.

Corollary 20. Let P be a Sylow p -subgroup of G . Then the following are equivalent:

- (1) P is the unique Sylow p -subgroup of G , i.e., $n_p = 1$
- (2) P is normal in G
- (3) P is characteristic in G
- (4) All subgroups generated by elements of p -power order are p -groups, i.e., if X is any subset of G such that $|x|$ is a power of p for all $x \in X$, then $\langle X \rangle$ is a p -group.

Proof: If (1) holds, then $gPg^{-1} = P$ for all $g \in G$ since $gPg^{-1} \in \text{Syl}_p(G)$, i.e., P is normal in G . Hence (1) implies (2). Conversely, if $P \trianglelefteq G$ and $Q \in \text{Syl}_p(G)$, then by Sylow's Theorem there exists $g \in G$ such that $Q = gPg^{-1} = P$. Thus $\text{Syl}_p(G) = \{P\}$ and (2) implies (1).

Since characteristic subgroups are normal, (3) implies (2). Conversely, if $P \trianglelefteq G$, we just proved P is the unique subgroup of G of order p^α , hence $P \text{ char } G$. Thus (2) and (3) are equivalent.

Finally, assume (1) holds and suppose X is a subset of G such that $|x|$ is a power of p for all $x \in X$. By the conjugacy part of Sylow's Theorem, for each $x \in X$ there is some $g \in G$ such that $x \in gPg^{-1} = P$. Thus $X \subseteq P$, and so $\langle X \rangle \leq P$, and $\langle X \rangle$ is a p -group. Conversely, if (4) holds, let X be the union of all Sylow p -subgroups of G . If P is any Sylow p -subgroup, P is a subgroup of the p -group $\langle X \rangle$. Since P is a p -subgroup of G of maximal order, we must have $P = \langle X \rangle$, so (1) holds.

Examples

Let G be a finite group and let p be a prime.

- (1) If p does not divide the order of G , the Sylow p -subgroup of G is the trivial group (and all parts of Sylow's Theorem hold trivially). If $|G| = p^\alpha$, G is the unique Sylow p -subgroup of G .
- (2) A finite abelian group has a unique Sylow p -subgroup for each prime p . This subgroup consists of all elements x whose order is a power of p . This is sometimes called the *p -primary component* of the abelian group.
- (3) S_3 has three Sylow 2-subgroups: $\langle (1\ 2) \rangle$, $\langle (2\ 3) \rangle$ and $\langle (1\ 3) \rangle$. It has a unique (hence normal) Sylow 3-subgroup: $\langle (1\ 2\ 3) \rangle = A_3$. Note that $3 \equiv 1 \pmod{2}$.
- (4) A_4 has a unique Sylow 2-subgroup: $\langle (1\ 2)(3\ 4), (1\ 3)(2\ 4) \rangle \cong V_4$. It has four Sylow 3-subgroups: $\langle (1\ 2\ 3) \rangle$, $\langle (1\ 2\ 4) \rangle$, $\langle (1\ 3\ 4) \rangle$ and $\langle (2\ 3\ 4) \rangle$. Note that $4 \equiv 1 \pmod{3}$.
- (5) S_4 has $n_2 = 3$ and $n_3 = 4$. Since S_4 contains a subgroup isomorphic to D_8 , every Sylow 2-subgroup of S_4 is isomorphic to D_8 .

Applications of Sylow's Theorem

We now give some applications of Sylow's Theorem. Most of the examples use Sylow's Theorem to prove that a group of a particular order is not simple. After discussing methods of constructing larger groups from smaller ones (for example, the formation of semidirect products) we shall be able to use these results to classify groups of some specific orders n (as we already did for $n = 15$).

Since Sylow's Theorem ensures the existence of p -subgroups of a finite group, it is worthwhile to study groups of prime power order more closely. This will be done in Chapter 6 and many more applications of Sylow's Theorem will be discussed there.

For groups of small order, the congruence condition of Sylow's Theorem alone is often sufficient to force the existence of a *normal* subgroup. The first step in any numerical application of Sylow's Theorem is to factor the group order into prime powers. The largest prime divisors of the group order tend to give the fewest possible values for n_p (for example, the congruence condition on n_2 gives no restriction whatsoever), which limits the structure of the group G . In the following examples we shall see situations where Sylow's Theorem alone does not force the existence of a normal subgroup, however some additional argument (often involving studying the elements of order p for a number of different primes p) proves the existence of a normal Sylow subgroup.

Example: (Groups of order pq , p and q primes with $p < q$)

Suppose $|G| = pq$ for primes p and q with $p < q$. Let $P \in \text{Syl}_p(G)$ and let $Q \in \text{Syl}_q(G)$. We show that Q is normal in G and if P is also normal in G , then G is cyclic.

Now the three conditions: $n_q = 1 + kq$ for some $k \geq 0$, n_q divides p and $p < q$, together force $k = 0$. Since $n_q = 1$, $Q \trianglelefteq G$.

Since n_p divides the prime q , the only possibilities are $n_p = 1$ or q . In particular, if $p \nmid q - 1$ (that is, if $q \not\equiv 1 \pmod{p}$), then n_p cannot equal q , so $P \trianglelefteq G$.

Let $P = \langle x \rangle$ and $Q = \langle y \rangle$. If $P \trianglelefteq G$, then since $G/C_G(P)$ is isomorphic to a subgroup of $\text{Aut}(Z_p)$ and the latter group has order $p - 1$, Lagrange's Theorem together with the observation that neither p nor q can divide $p - 1$ implies that $G = C_G(P)$. In this case $x \in P \leq Z(G)$ so x and y commute. (Alternatively, this follows immediately from Exercise 42 of Section 3.1.) This means $|xy| = pq$ (cf. the exercises in Section 2.3), hence in this case G is cyclic: $G \cong Z_{pq}$.

If $p \mid q - 1$, we shall see in Chapter 5 that there is a unique non-abelian group of order pq (in which, necessarily, $n_p = q$). We can prove the existence of this group now. Let Q be a Sylow q -subgroup of the symmetric group of degree q , S_q . By Exercise 34 in Section 3, $|N_{S_q}(Q)| = q(q - 1)$. By assumption, $p \mid q - 1$ so by Cauchy's Theorem $N_{S_q}(Q)$ has a subgroup, P , of order p . By Corollary 15 in Section 3.2, PQ is a group of order pq . Since $C_{S_q}(Q) = Q$ (Example 2, Section 3), PQ is a non-abelian group. The essential ingredient in the uniqueness proof of PQ is Theorem 17 on the cyclicity of $\text{Aut}(Z_q)$.

Example: (Groups of order 30)

Let G be a group of order 30. We show that G has a normal subgroup isomorphic to Z_{15} . We shall use this information to classify groups of order 30 in the next chapter. Note that any subgroup of order 15 is necessarily normal (since it is of index 2) and cyclic (by the preceding result) so it is only necessary to show there exists a subgroup of order 15. The quickest way of doing this is to quote Exercise 13 in Section 2. We give an alternate argument which illustrates how Sylow's Theorem can be used in conjunction with a counting of elements of prime order to produce a normal subgroup.

Let $P \in \text{Syl}_5(G)$ and let $Q \in \text{Syl}_3(G)$. If either P or Q is normal in G , by Corollary 15, Chapter 3, PQ is a group of order 15. Note also that if either P or Q is normal, then both P and Q are characteristic subgroups of PQ , and since $PQ \trianglelefteq G$, both P and Q are normal in G (Exercise 8(a), Section 4). Assume therefore that neither Sylow subgroup is normal. The only possibilities by Part 3 of Sylow's Theorem are $n_5 = 6$ and $n_3 = 10$. Each element of order 5 lies in a Sylow 5-subgroup, each Sylow 5-subgroup contains 4 nonidentity elements and, by Lagrange's Theorem, distinct Sylow 5-subgroups intersect in the identity. Thus the number of elements of order 5 in G is the number of nonidentity elements in one Sylow 5-subgroup times the number of Sylow 5-subgroups. This would

be $4 \cdot 6 = 24$ elements of order 5. By similar reasoning, the number of elements of order 3 would be $2 \cdot 10 = 20$. This is absurd since a group of order 30 cannot contain $24 + 20 = 44$ distinct elements. One of P or Q (hence both) must be normal in G .

This sort of counting technique is frequently useful (cf. also Section 6.2) and works particularly well when the Sylow p -subgroups have order p (as in this example), since then the intersection of two distinct Sylow p -subgroups must be the identity. If the order of the Sylow p -subgroup is p^α with $\alpha \geq 2$, greater care is required in counting elements, since in this case distinct Sylow p -subgroups may have many more elements in common, i.e., the intersection may be nontrivial.

Example: (Groups of order 12)

Let G be a group of order 12. We show that either G has a normal Sylow 3-subgroup or $G \cong A_4$ (in the latter case G has a normal Sylow 2-subgroup). We shall use this information to classify groups of order 12 in the next chapter.

Suppose $n_3 \neq 1$ and let $P \in \text{Syl}_3(G)$. Since $n_3 \mid 4$ and $n_3 \equiv 1 \pmod{3}$, it follows that $n_3 = 4$. Since distinct Sylow 3-subgroups intersect in the identity and each contains two elements of order 3, G contains $2 \cdot 4 = 8$ elements of order 3. Since $|G : N_G(P)| = n_3 = 4$, $N_G(P) = P$. Now G acts by conjugation on its four Sylow 3-subgroups, so this action affords a permutation representation

$$\varphi : G \rightarrow S_4$$

(note that we could also act by left multiplication on the left cosets of P and use Theorem 3). The kernel K of this action is the subgroup of G which normalizes all Sylow 3-subgroups of G . In particular, $K \leq N_G(P) = P$. Since P is not normal in G by assumption, $K = 1$, i.e., φ is injective and

$$G \cong \varphi(G) \leq S_4.$$

Since G contains 8 elements of order 3 and there are precisely 8 elements of order 3 in S_4 , all contained in A_4 , it follows that $\varphi(G)$ intersects A_4 in a subgroup of order at least 8. Since both groups have order 12 it follows that $\varphi(G) = A_4$, so that $G \cong A_4$.

Note that A_4 does indeed have 4 Sylow 3-subgroups (see Example 4 following Corollary 20), so that such a group G does exist. Also, let V be a Sylow 2-subgroup of A_4 . Since $|V| = 4$, it contains all of the remaining elements of A_4 . In particular, there cannot be another Sylow 2-subgroup. Thus $n_2(A_4) = 1$, i.e., $V \trianglelefteq A_4$ (which one can also see directly because V is the identity together with the three elements of S_4 which are products of two disjoint transpositions, that is, V is a union of conjugacy classes).

Example: (Groups of order p^2q , p and q distinct primes)

Let G be a group of order p^2q . We show that G has a normal Sylow subgroup (for either p or q). We shall use this information to classify some groups of this order in the next chapter (cf. Exercises 8 to 12 of Section 5.5). Let $P \in \text{Syl}_p(G)$ and let $Q \in \text{Syl}_q(G)$.

Consider first when $p > q$. Since $n_p \mid q$ and $n_p = 1 + kp$, we must have $n_p = 1$. Thus $P \trianglelefteq G$.

Consider now the case $p < q$. If $n_q = 1$, Q is normal in G . Assume therefore that $n_q > 1$, i.e., $n_q = 1 + tq$, for some $t > 0$. Now n_q divides p^2 so $n_q = p$ or p^2 . Since $q > p$ we cannot have $n_q = p$, hence $n_q = p^2$. Thus

$$tq = p^2 - 1 = (p - 1)(p + 1).$$

Since q is prime, either $q \mid p - 1$ or $q \mid p + 1$. The former is impossible since $q > p$ so the latter holds. Since $q > p$ but $q \mid p + 1$, we must have $q = p + 1$. This forces $p = 2$, $q = 3$ and $|G| = 12$. The result now follows from the preceding example.

Groups of Order 60

We illustrate how Sylow's Theorems can be used to unravel the structure of groups of a given order even if some groups of that order may be simple. Note the technique of changing from one prime to another and the inductive process where we use results on groups of order < 60 to study groups of order 60.

Proposition 21. If $|G| = 60$ and G has more than one Sylow 5-subgroup, then G is simple.

Proof: Suppose by way of contradiction that $|G| = 60$ and $n_5 > 1$ but that there exists H a normal subgroup of G with $H \neq 1$ or G . By Sylow's Theorem the only possibility for n_5 is 6. Let $P \in \text{Syl}_5(G)$, so that $|N_G(P)| = 10$ since its index is n_5 .

If $5 \mid |H|$ then H contains a Sylow 5-subgroup of G and since H is normal, it contains all 6 conjugates of this subgroup. In particular, $|H| \geq 1 + 6 \cdot 4 = 25$, and the only possibility is $|H| = 30$. This leads to a contradiction since a previous example proved that any group of order 30 has a normal (hence unique) Sylow 5-subgroup. This argument shows 5 does not divide $|H|$ for any proper normal subgroup H of G .

If $|H| = 6$ or 12, H has a normal, hence characteristic, Sylow subgroup, which is therefore also normal in G . Replacing H by this subgroup if necessary, we may assume $|H| = 2, 3$ or 4. Let $\bar{G} = G/H$, so $|\bar{G}| = 30, 20$ or 15. In each case, $\bar{G}$ has a normal subgroup $\bar{P}$ of order 5 by previous results. If we let H_1 be the complete preimage of $\bar{P}$ in G , then $H_1 \trianglelefteq G$, $H_1 \neq G$ and $5 \mid |H_1|$. This contradicts the preceding paragraph and so completes the proof.

Corollary 22. A_5 is simple.

Proof: The subgroups $\langle (1\ 2\ 3\ 4\ 5) \rangle$ and $\langle (1\ 3\ 2\ 4\ 5) \rangle$ are distinct Sylow 5-subgroups of A_5 so the result follows immediately from the proposition.

The next proposition shows that there is a unique simple group of order 60.

Proposition 23. If G is a simple group of order 60, then $G \cong A_5$.

Proof: Let G be a simple group of order 60, so $n_2 = 3, 5$ or 15. Let $P \in \text{Syl}_2(G)$ and let $N = N_G(P)$, so $|G : N| = n_2$.

First observe that G has no proper subgroup H of index less than 5, as follows: if H were a subgroup of G of index 4, 3 or 2, then, by Theorem 3, G would have a normal subgroup K contained in H with G/K isomorphic to a subgroup of S_4 , S_3 or S_2 . Since $K \neq G$, simplicity forces $K = 1$. This is impossible since $60 (= |G|)$ does not divide $4!$. This argument shows, in particular, that $n_2 \neq 3$.

If $n_2 = 5$, then N has index 5 in G so the action of G by left multiplication on the set of left cosets of N gives a permutation representation of G into S_5 . Since (as

above) the kernel of this representation is a proper normal subgroup and G is simple, the kernel is 1 and G is isomorphic to a subgroup of S_5 . Identify G with this isomorphic copy so that we may assume $G \leq S_5$. If G is not contained in A_5 , then $S_5 = GA_5$ and, by the Second Isomorphism Theorem, $A_5 \cap G$ is of index 2 in G . Since G has no (normal) subgroup of index 2, this is a contradiction. This argument proves $G \leq A_5$. Since $|G| = |A_5|$, the isomorphic copy of G in S_5 coincides with A_5 , as desired.

Finally, assume $n_2 = 15$. If for every pair of distinct Sylow 2-subgroups P and Q of G , $P \cap Q = 1$, then the number of nonidentity elements in Sylow 2-subgroups of G would be $(4 - 1) \cdot 15 = 45$. But $n_5 = 6$ so the number of elements of order 5 in G is $(5 - 1) \cdot 6 = 24$, accounting for 69 elements. This contradiction proves that there exist distinct Sylow 2-subgroups P and Q with $|P \cap Q| = 2$. Let $M = N_G(P \cap Q)$. Since P and Q are abelian (being groups of order 4), P and Q are subgroups of M and since G is simple, $M \neq G$. Thus 4 divides $|M|$ and $|M| > 4$ (otherwise, $P = M = Q$). The only possibility is $|M| = 12$, i.e., M has index 5 in G (recall M cannot have index 3 or 1). But now the argument of the preceding paragraph applied to M in place of N gives $G \cong A_5$. This leads to a contradiction in this case because $n_2(A_5) = 5$ (cf. the exercises). The proof is complete.

EXERCISES

Let G be a finite group and let p be a prime.

1. Prove that if $P \in \text{Syl}_p(G)$ and H is a subgroup of G containing P then $P \in \text{Syl}_p(H)$. Give an example to show that, in general, a Sylow p -subgroup of a subgroup of G need not be a Sylow p -subgroup of G .
2. Prove that if H is a subgroup of G and $Q \in \text{Syl}_p(H)$ then $gQg^{-1} \in \text{Syl}_p(gHg^{-1})$ for all $g \in G$.
3. Use Sylow's Theorem to prove Cauchy's Theorem. (Note that we only used Cauchy's Theorem for abelian groups — Proposition 3.21 — in the proof of Sylow's Theorem so this line of reasoning is not circular.)
4. Exhibit all Sylow 2-subgroups and Sylow 3-subgroups of D_{12} and $S_3 \times S_3$.
5. Show that a Sylow p -subgroup of D_{2n} is cyclic and normal for every odd prime p .
6. Exhibit all Sylow 3-subgroups of A_4 and all Sylow 3-subgroups of S_4 .
7. Exhibit all Sylow 2-subgroups of S_4 and find elements of S_4 which conjugate one of these into each of the others.
8. Exhibit two distinct Sylow 2-subgroups of S_5 and an element of S_5 that conjugates one into the other.
9. Exhibit all Sylow 3-subgroups of $SL_2(\mathbb{F}_3)$ (cf. Exercise 9, Section 2.1).
10. Prove that the subgroup of $SL_2(\mathbb{F}_3)$ generated by $\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}$ and $\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ is the unique Sylow 2-subgroup of $SL_2(\mathbb{F}_3)$ (cf. Exercise 10, Section 2.4).
11. Show that the center of $SL_2(\mathbb{F}_3)$ is the group of order 2 consisting of $\pm I$, where I is the identity matrix. Prove that $SL_2(\mathbb{F}_3)/Z(SL_2(\mathbb{F}_3)) \cong A_4$. [Use facts about groups of order 12.]
12. Let $2n = 2^a k$ where k is odd. Prove that the number of Sylow 2-subgroups of D_{2n} is k . [Prove that if $P \in \text{Syl}_2(D_{2n})$ then $N_{D_{2n}}(P) = P$.]